Exploratory Q-Network: A DRL Framework for MOP-Based GB-SAR Deployment in Open-Pit Coal Mine Slope Monitoring

Weiming Li, Zhiqiang Du, and Heng Zhang

***Abstract*—Optimal ground-based synthetic aperture radar (GB-SAR) deployment is essential for high-precision slope monitoring in open-pit mines. Existing studies typically formulate GB-SAR deployment as a multiobjective optimization problem (MOP) solved by deep reinforcement learning (DRL). However, these approaches are prone to local optima in high-dimensional spaces due to three limitations: 1) the random exploration of conventional ϵ-greedy strategies is inefficient for discovering experiences with higher rewards; 2) purely objective-driven rewards fail to motivate novel state exploration; and 3) gradient dilution in high-dimensional action spaces impedes stable convergence. To overcome these challenges, we propose the exploratory Q-network (EQN). It incorporates three components: 1) a multihead action confluence (MHAC) module that decomposes and dynamically aggregates action advantages; 2) state uncertainty rewards (SURs) with an anti-forgetting design to escape local optima and mitigate forgetting; and 3) policy perturbation factors (PPFs) that adopt NoisyNet-style perturbations to steer exploration toward higher reward paths. Multiperiod experiments at an open-pit coal mine demonstrate the efficacy of EQN, achieving 80.09% coverage and 74.95% coherence. At a 6-m resolution, EQN improves coherence by 6.16% and 12.39% and coverage by 1.34% and 19.13% when compared to the double dueling deep Q-network (D3QN) and Nondominated Sorting Genetic Algorithm II (NSGA-II), respectively.**

***Index Terms*—Deep reinforcement learning (DRL), ground-based synthetic aperture radar (GB-SAR) deployment, mitigating local optima, multiobjective optimization, slope monitoring.**

I. Introduction

OPTIMIZING ground-based synthetic aperture radar (GB-SAR) deployment in open-pit coal mine slopes is pivotal for enhancing slope monitoring capability. Open-pit coal mine slope failures occur frequently worldwide and result in significant losses, posing a substantial threat to mine production [1], [2]. GB-SAR plays a crucial role in current monitoring technologies. The advantages of all-weather operation, large-area observation, and high-precision interferometric measurement position it as an effective tool for monitoring open-pit coal mine slopes [3], [4], [5]. The monitoring efficacy of GB-SAR is highly sensitive to its deployment configuration. Minor deviations in the device's position or orientation can result in significant degradation of data quality, particularly in terms of coherence and spatial coverage [6]. Currently, deployment largely relies on manual experience, which is labor-intensive and often yields suboptimal solutions, thereby failing to fully exploit the hardware's potential. Approaches based on artificial intelligence (AI) have demonstrated considerable potential in addressing such challenges recently. Therefore, optimizing GB-SAR deployment through AI-driven approaches is crucial to maximize monitoring efficacy.

The deployment of GB-SAR essentially constitutes a multiobjective optimization problem (MOP), necessitating a tradeoff between conflicting objectives such as coverage, measurement accuracy, and operational costs. Traditional swarm-based or evolutionary algorithms [7] and deep reinforcement learning (DRL) methods [8] are commonly employed to address MOPs. However, these methods face the challenge of being prone to being trapped in local optima in high-dimensional spaces. The GB-SAR deployment scenario is characterized by a discrete and high-dimensional space, where traditional methods suffer from the curse of dimensionality, resulting in reduced solution diversity and monitoring efficiency [9], as illustrated in Fig. 1(a). DRL methods hold greater potential than traditional methods to find superior solutions in these spaces. For instance, deep Q-network (DQN) and its variants [10], [11], [12] can find better solutions by learning strategies through the interactions between neural network-based agents and the environment [13], but they remain susceptible to being trapped in local optima when facing even higher dimensional state-action spaces. Three primary limitations in current DRL methods contribute to this failure.

1) The standard ϵ-greedy exploration strategy relies on random sampling for exploration. The probability of stumbling upon high-value experiences is negligible, limiting the agent's ability to escape local optima.
2) Agents lack intrinsic motivation to explore novel states because they rely entirely on reward signals provided by the objective functions. Consequently, agents tend to converge prematurely to known local optima.
3) Learning signals become severely diluted as the number of network parameters required to estimate Q values

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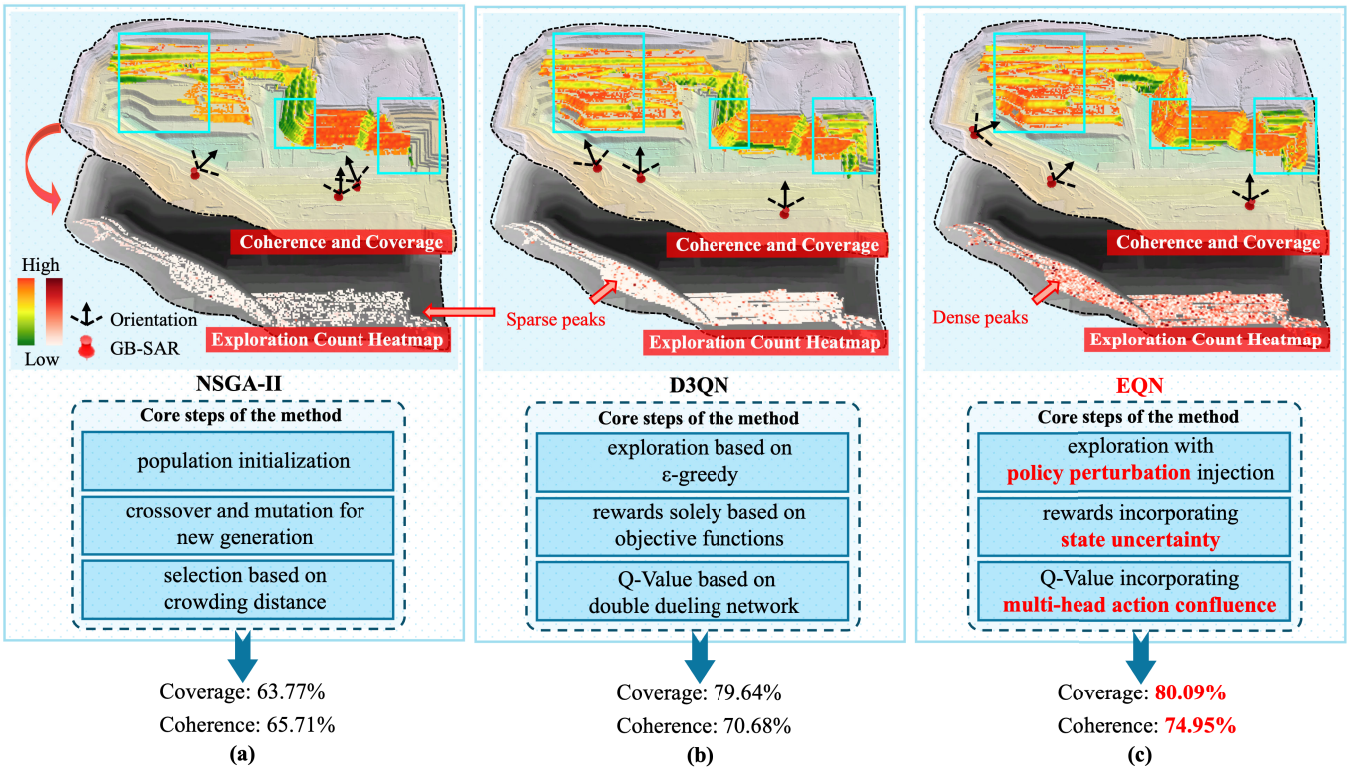


Fig. 1. Deployment results of three optimization methods with the 7.5-m resolution DSM. (a) Deployment results of NSGA-II. The upper figure shows the monitoring coverage area of GB-SAR on the slope and the distribution of coherence, where red indicates high coherence and green indicates low coherence. The lower figure is the exploration count heatmap. Each cell value represents the number of visits by the agent during the training phase at that location, with colors ranging from white to red indicating increasing visit frequency. The heatmap is used to evaluate the agent's exploration ability. Sparse red peaks indicate that the agent is trapped in local optima. (b) Deployment results of D3QN. (c) Deployment results of EQN. Higher coherence and coverage, along with dense peak distributions, indicate that the local optima problem has been alleviated. In addition, the areas marked with blue boxes indicate where the three methods primarily differ.

explodes. As a result, updates for numerous parameters lag behind, causing persistent oscillations of the network during training.

Recent studies have attempted to enhance exploration and reward strategies, yet limitations persist. Approaches such as adaptive ϵ-greedy variants [14], [15], [16] adjust exploration rates but technically remain dependent on random selection. The probability of discovering high-value experiences through such nondirectional exploration is negligible in high-dimensional space. Intrinsic reward mechanisms, including count-based methods [17] and forward dynamics models [18], [19], offer improvements but suffer from high storage overhead or catastrophic forgetting issues. Furthermore, decomposition methods are applied to reduce the dimensions of high-dimensional discrete action spaces [20]. Weight assignment is challenging because the relative importance of decomposed subactions varies dynamically with state transitions. Although the aforementioned methods have achieved certain progress, their respective limitations remain significant, and few studies have successfully integrated directed exploration, intrinsic motivation, and action space decomposition into a unified system for more powerful optimization capabilities.

To bridge this gap, this article proposes a GB-SAR deployment framework named exploratory Q-network (EQN) tailored for mitigating local optima problems in high-dimensional state-action spaces, as shown in Fig. 1(c). An MOP for GB-SAR deployment is first established in this article to optimize interferometric deformation accuracy and coverage. The MOP is then decomposed into scalar subproblems. Second, the proposed EQN serves to solve each subproblem, ultimately generating the Pareto front. Specifically, a multihead action confluence (MHAC) module is proposed. This module decomposes the action advantages within the dueling architecture into multiple branches and dynamically aggregates the advantage values from these branches, thereby reducing the dimensions of the action space. A state prediction network (SPN) and a state target network (STN) are also proposed. These networks provide the agent with state uncertainty rewards (SURs), motivating it to explore novel states. Meanwhile, an anti-forgetting network (AFN) is designed to mitigate the SPN's forgetting problem. Policy perturbation factors (PPFs) that adopt NoisyNet-style perturbations [21] are injected into the Q-network, empowering the agent to deliberately execute actions leading to novel paths. Finally, experiments on open-pit coal mine slopes are conducted. The main contributions of this article are summarized as follows.

1) An MOP is formulated for GB-SAR deployment to balance interferometric deformation accuracy and coverage while fully incorporating the specific observation characteristics of GB-SAR.

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2) An MHAC module with a dynamic weight generator (DWG) is designed, which decomposes the action space into branches to enable EQN to converge stably in high-dimensional spaces and aggregates them by their contributions to make accurate decisions.
3) SURs with an anti-forgetting design are developed, which incentivize the agent to escape local optima and mitigate model forgetting throughout training.

The rest of this article is organized as follows. Section II reviews the related work. Section III details the proposed methodology. Section IV presents the experiments and discusses the results. Section V concludes this article and outlines future work.

II. Related Work

The MOP of GB-SAR deployment aims to find the optimal solution that balances conflicting objectives, primarily monitoring accuracy, spatial coverage, and economic constraints. A clear definition of these performance indicators is needed to formulate the MOP. Various traditional optimization methods have been adopted to address this MOP, including swarm-based and evolutionary algorithms. Moreover, the recent rapid development of DRL enhances its potential in handling complex decision-making and high-dimensional optimization problems, positioning it as a powerful tool for solving this MOP.

A. Objective Function Formulation

The objective function guides the algorithm's optimization process to progressively acquire Pareto optimal solutions. In the context of GB-SAR, preserving the quality of differential interferometric measurements is paramount. While the ultimate goal is precise deformation retrieval, the coherence coefficient acts as a critical proxy for evaluating interferogram quality, as high coherence correlates directly with reduced phase noise and accurate phase unwrapping [22], [23]. Consequently, modeling coherence is central to the optimization objective. Some approaches compute coherence by performing complex multiplication on consecutive interferograms [24], [25], [26]. These methods require computationally expensive simulations, which are unsuitable for iterative optimization processes. The correlation coefficient synthesis method enhances the spatio-temporal reliability of correlation maps by integrating various decoherence factors such as volume, time, and thermal noise [27], [28]. These approaches can reduce the computational effort required to obtain coherence. In addition, studies have also focused on the monitoring coverage and construction costs. The coverage objective function ensures observation of critical areas, while the construction cost objective function imposes practical budget constraints [29].

B. Traditional and DRL-Based Methods for MOP

1) Traditional Methods: Evolutionary and swarm intelligence algorithms have been the dominant solvers for GB-SAR deployment. Standard approaches like the Nondominated Sorting Genetic Algorithm II (NSGA-II) have been adapted with specific constraints to filter infeasible solutions [24]. The multiobjective cooperative particle swarm optimization (MOCPSO) algorithm integrates both diversity and convergence search strategies to improve solution quality [30]. Studies have also been conducted in the field of multiobjective evolutionary algorithms (MOEAs). The MOEA based on hierarchical decomposition (MOEA/HD) decomposes the multiobjective problem into simpler subproblems and coordinates their optimization to improve search efficiency [31]. The MOEA/HD with a weighted optimization framework (MOEA/D-WOF) employs Gaussian process models to predict high-quality solutions [32]. Although these algorithms have improved the performance of solving MOPs, they face challenges in high-dimensional spaces. Population updates in swarm-based methods find it difficult to escape local optima accidentally discovered by individuals, and the crossover and mutation operations in evolutionary methods struggle to reorganize beneficial genetic information [33].

2) DRL-Based Methods: DRL methods combine the capabilities of neural networks with the interactive optimization framework of reinforcement learning to improve the quality of the Pareto front. The GB-SAR deployment space is treated as discretized to reduce the problem scale. Value-based DRL methods are frequently applied to such problems because of their suitability for discrete state-action spaces [34], [35], [36]. Researchers model the MOP as Markov decision process (MDP) and utilize DQN to learn the optimal policies [37]. Researchers further construct an architecture that integrates the strengths of various Q-network models, namely the prioritized experience replay-double dueling DQN (PER-D3QN) [38], [39], [40]. It aims at enhancing the algorithm's adaptability in complex decision spaces. The distributed DQN method treats each deployment object in the system as an independent agent [41]. Each agent independently processes a portion of the states and makes decisions, which reduces the enormous computational overhead caused by processing global states. The linear scalarized multiobjective DQN (LSMO-DQN) employs the linear scalarization method to transform MOP into single-objective problems for solving [42], accelerating the acquisition speed of the Pareto front.

It is noted that traditional methods exhibit shortcomings in the quality of the generated Pareto fronts and the monitoring efficiency in high-dimensional spaces. Although DRL techniques have partially addressed limitations of traditional methods, their widely adopted random exploration strategies struggle to find high-value experiences in high-dimensional spaces. Agents also rely excessively on sparse external rewards and lack intrinsic motivation. The exponential growth of the action space further challenges the network's convergence capability in high-dimensional spaces. This article proposes PPF and SUR to enhance the agent's exploration and reward strategies, and designs MHAC to reduce the dimensions of the action space.

C. Methods for Addressing High-Dimensional Spaces

Several advanced strategies have been proposed to mitigate the limitations of the aforementioned DRL methods in high-dimensional spaces, yet each possesses inherent drawbacks.

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The preference branch is introduced to learn the action preference distribution [43], which enables the agent to sample actions with probability ϵ based on the learned distribution. This strategy replaces the purely random exploration in traditional ϵ-greedy strategies, making exploration more targeted. However, the sampling process lacks random perturbation, making the probability distribution prone to excessive concentration and causing exploration to focus on certain actions. Other studies have integrated the Tanh activation function to adjust the ϵ value automatically based on the number of training episodes [44]. Although the adaptive adjustment of ϵ helps to better balance exploration and exploitation, the exploration phase still executes actions randomly and inherently remains devoid of specific directionality. NoisyNet [21] instead injects learnable parametric noise into the network weights, yielding state-dependent and self-adaptive exploration that supersedes the undirected ϵ-greedy strategy. Some studies leverage forward dynamics models to predict the next state after an agent performs an action [45], [46]. They treat the error between the predicted next state and the actual next state as an additional reward to incentivize the agent to explore unfamiliar states. Such methods overlook the issue of model forgetting during training, which can lead to unstable rewards. Hierarchical methods decompose high-dimensional action spaces into a series of nested decision tiers [47], [48]. Higher decision levels make coarse-grained selections, while lower decision levels handle finer-grained actions. These approaches encounter challenges in coordinating between high-level and low-level policies. Branching methods decompose actions into multiple branches [49]. Although they can reduce the dimensions of the action space handled by each branch, the outputs of branches are aggregated through simple averaging or similar techniques and the relative contributions of each branch are neglected, leading to inaccurate optimization results.

This article employs the PPF to fundamentally transform exploration into a directed process by injecting the learnable perturbations introduced by NoisyNet into the policy layer. The SUR is designed to provide intrinsic rewards while mitigating the issue of forgetting across the entire training process. The MHAC module is constructed to decompose actions into parallel subaction branches and subsequently aggregates Q-values via a DWG. These modules are fused together to enhance the ability to mitigate local optima in high-dimensional space.

III. Method Framework

This section focuses on detailing the construction of the GB-SAR deployment space, the formulation of the MOP, and the EQN-based MOP solution method.

A. Construction of GB-SAR Deployment Space

The deployment space construction aims to transform complex open-pit coal mine slopes into a discretized and structured space where each unit carries critical attributes. Fig. 2(a) illustrates the diagram of the GB-SAR deployment space. The digital surface model (DSM) of the open-pit coal mine is first acquired via autonomous aerial vehicle photogrammetry. Target slope regions are manually delineated based on the DSM. Then, a DSM-based slope analysis is performed to identify flat deployment regions. Engineering practice sets a slope threshold of 10° to exclude grids with excessive slopes and retain only flat areas. Subsequently, a line-of-sight (LOS) analysis is conducted to evaluate potential occlusions between candidate grids and the slope grids, which further reduces the grid count and problem dimension. Specifically, the analysis determines whether an obstruction exists between two grids by calculating the elevation of interpolated points along the line of sight using bilinear interpolation [50]. The density of interpolation points is adapted to the terrain complexity. Candidate grids are excluded if their observation rate for the entire slope, defined as the ratio of visible slope grids to the total number of slope grids, falls below a preset threshold, typically 10%. The centers of these retained grids are considered as candidate deployment positions for the GB-SAR.

B. Formulation of MOP for GB-SAR Deployment

Coverage directly determines the GB-SAR's observational capability over the slope area. The optimization of GB-SAR coverage aims to ensure comprehensive acquisition of deformation information across the entire slope. The set of observable slope grids for each GB-SAR is determined using the LOS analysis method described before, considering the constraints of maximum horizontal and vertical observation angles, as well as maximum and minimum measurable ranges. The slope coverage objective function for the deployment is defined as follows:

$$F_{\text{cover}} = \frac{|G_{\text{visible}}|}{|G_{\text{total}}|} \tag{1}$$

where G_{visible} represents the set of all slope grids that can be observed by all deployed devices and G_{total} is the total number of slope grids.

The accuracy of GB-SAR's differential interferometric measurements directly depends on the quality of the interferogram. To ensure high-precision deformation retrieval, the coherence coefficient is employed as the objective function. High coherence signifies lower phase noise in the interferogram. The lower phase noise reduces phase unwrapping errors and enables accurate extraction of true deformation phase information. This objective function for the deployment is defined by synthesizing individual coherence components [51] as follows:

$$F_{\text{coherence}} = \text{Median}\left\{\gamma_j \mid j \in G_{\text{visible}}\right\} \tag{2}$$

$$\gamma_j = \max_{i \in D_{\text{visible}}(j)} \gamma_j(i) \tag{3}$$

$$\gamma_j(i) = \gamma_{\text{geom}} \cdot \gamma_{\text{DC}} \cdot \gamma_{\text{temp}} \cdot \gamma_{\text{proc}} \cdot \gamma_{\text{ther},ij} \cdot \gamma_{\text{vol},ij} \tag{4}$$

where $\gamma_j(i)$ is the coherence of the *jth* slope grid observed by the *ith* GB-SAR, $D_{\text{visible}}(j)$ denotes the set of deployed devices capable of observing this grid, and γ_j represents the maximum coherence for the *jth* slope grid achieved by one of these devices.

The geometric decoherence γ_{geom} results from the spatial baseline length between two adjacent observations. The Doppler decoherence γ_{DC} arises from differences in the radar's

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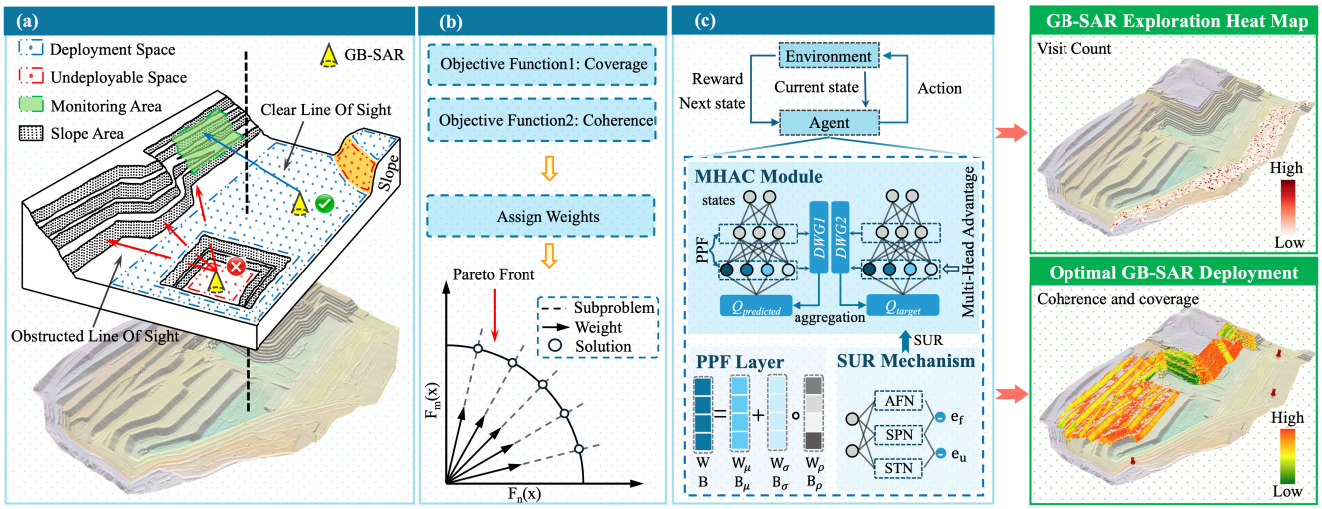


Fig. 2. Overall flowchart of the proposed method for deploying GB-SAR in the open-pit coal mine. (a) Construction of GB-SAR deployment space. (b) Formulation of MOP. (c) Solving of MOP based on EQN.

Doppler center frequency during adjacent observations. The temporal decoherence γ_{temp} comes from changes in the scene's physical properties during the observation interval. The processing decoherence γ_{proc} originates from errors during data processing. Practical deployment optimization of GB-SAR generally treats these four decoherences as constants. GB-SAR typically operates in a repeat-pass mode, with observations taken from nearly the same location, which minimizes γ_{geom} and γ_{DC}. The temporal decoherence γ_{temp} mainly depends on the monitoring cycle. The processing decoherence γ_{proc} relies on the precision of the postprocessing algorithm. Neither γ_{temp} nor γ_{proc} depends on the deployment status. The primary focus lies on thermal noise decoherence and volumetric decoherence since these two are most significantly affected by the deployment position and orientation.

Thermal noise decoherence describes the loss of coherence caused by the impact of internal system thermal noise on the signal-to-noise ratio (SNR). It is defined as follows:

$$\gamma_{\text{ther},ij} = \frac{SNR}{1+SNR} = \frac{PR_{ij}}{PN+PR_{ij}} \tag{5}$$

where PN represents the noise power and PR_{ij} represents the radar return signal power that is defined by the radar equation

$$PR_{ij} = \frac{P_t G_t G_r \sigma_{ij} \lambda^2}{(4\pi)^3 d_{ij}^4 L} \tag{6}$$

where P_t denotes the transmitted power, G_t and G_r denote the antenna gains, λ denotes the radar wavelength, and L denotes the system loss factor. d_{ij} is the distance between the *ith* GB-SAR and the *jth* slope grid.

The radar cross section σ_{ij} of the *jth* slope grid observed by the *ith* GB-SAR is obtained by integrating the slope's backscattering coefficient σ_{ij}^0 [52] over its area A_j

$$\sigma_{ij} = \sigma_{ij}^0 \cdot A_j = g\sqrt{p}\cos^3\vartheta_{ij}\left[\Gamma_{VV}\left(\vartheta_{ij}\right) + \Gamma_{HH}\left(\vartheta_{ij}\right)\right] \tag{7}$$

where g represents the soil roughness and $\sqrt{p}$ represents the penetration depth. Γ_{VV} and Γ_{HH} are the Fresnel reflection coefficients. ϑ_{ij} is the incidence angle. The model in [52] is calibrated under a far-field, plane-wave scattering condition that is satisfied by the GB-SAR observation geometry. The far-field distance is governed by the physical antenna dimension d through $R_{\text{ff}} = 2d^2/\lambda$, rather than the synthetic aperture, which is formed in postprocessing and never physically radiates. With a submeter physical antenna and centimeter-scale wavelength, R_{ff} stays on the order of meters. Since typical open-pit mine observation ranges exceed this, wavefront curvature within each resolution grid is negligible and the illumination can be treated as a local plane wave. g is defined as follows:

$$g = 0.7\left[1 - \exp\left(-0.65\left(\frac{2\pi s}{\lambda}\right)^{1.8}\right)\right] \tag{8}$$

where s denotes the soil surface root mean square height. $\sqrt{p}$ is defined as follows:

$$\sqrt{p} = 1 - \left(\frac{2\vartheta_{ij}}{\pi}\right)^{\frac{1}{3\Gamma_0}} \cdot \exp\left(-\frac{2\pi s}{\lambda}\right) \tag{9}$$

where $\Gamma_0 = |(1-(\epsilon_r)^{1/2})/(1+(\epsilon_r)^{1/2})|^2$. ϵ_r is the soil complex dielectric constant, typically expressed as $\epsilon_r = 3.03 + 9.3\ \text{m}_v + 146.0\ \text{m}_v^2 - 76.7\ \text{m}_v^3 - j\cdot 2.996\lambda$ [53]. m_v is the soil volumetric water content. $\Gamma_{VV}(\vartheta_{ij})$ and $\Gamma_{HH}(\vartheta_{ij})$ are defined as follows:

$$\Gamma_{VV}\left(\vartheta_{ij}\right) = \left|\frac{\epsilon_r\cos\vartheta_{ij} - \sqrt{\epsilon_r - \sin^2\vartheta_{ij}}}{\epsilon_r\cos\vartheta_{ij} + \sqrt{\epsilon_r - \sin^2\vartheta_{ij}}}\right|^2 \tag{10}$$

$$\Gamma_{HH}\left(\vartheta_{ij}\right) = \left|\frac{\cos\vartheta_{ij} - \sqrt{\epsilon_r - \sin^2\vartheta_{ij}}}{\cos\vartheta_{ij} + \sqrt{\epsilon_r - \sin^2\vartheta_{ij}}}\right|^2. \tag{11}$$

ϑ_{ij} represents the angle between the *ith* GB-SAR's LOS direction and the normal to the *jth* slope surface. It is defined as follows:

$$\vartheta_{ij} = \frac{\min\left[\arccos\left(\hat{r_{ij}}\cdot\hat{n_j}\right), 180^\circ - \arccos\left(\hat{r_{ij}}\cdot\hat{n_j}\right)\right]}{90^\circ} \tag{12}$$

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where $\hat{r_{ij}}$ is the unit vector from the *ith* GB-SAR to the *jth* slope grid and $\hat{n_j}$ is the unit surface normal vector at the *jth* slope grid. They can be calculated as follows:

$$\hat{r_{ij}} = \left(\frac{x_i - x_j}{d_{ij}}, \frac{y_i - y_j}{d_{ij}}, \frac{z_i - z_j}{d_{ij}} \right) \tag{13}$$

$$\hat{n_j} = \left(\frac{-g_x}{\sqrt{g_x^2 + g_y^2 + 1}}, \frac{-g_y}{\sqrt{g_x^2 + g_y^2 + 1}}, \frac{1}{\sqrt{g_x^2 + g_y^2 + 1}} \right) \tag{14}$$

where (x_i, y_i, z_i) represents the coordinates of the *ith* device and (x_j, y_j, z_j) represents the coordinates of the *jth* slope grid center. The elevation gradients g_x and g_y in *X*- and *Y*-directions, respectively, are computed using the Sobel operator [54].

Volumetric decoherence is caused by the penetration and scattering of the radar wave within the scatterer. It is defined as follows:

$$\gamma_{\text{vol},ij} = \frac{1}{\sqrt{1 + \left(\frac{2\pi \sqrt{\epsilon_r'} p B_\perp}{d_{ij} \lambda \tan \vartheta_{ij}} \right)^2}} \tag{15}$$

where $\epsilon_r' = 3.03 + 9.3\ \text{m}_v + 146.0\ \text{m}_v^2 - 76.7\ \text{m}_v^3$ and $B_\perp$ denotes the perpendicular baseline.

The objectives are then normalized. F_{cover} inherently represents a percentage and falls within $[0, 1]$. To align $F_{coherence}$ with the coverage scale, γ_j is normalized using Min-Max scaling as follows:

$$\tilde{\gamma}_j = \frac{\gamma_j - \gamma_{\min}}{\gamma_{\max} - \gamma_{\min}} \tag{16}$$

where $\gamma_{\max}$ and $\gamma_{\min}$ are the global maximum and minimum coherence values across the deployment space, respectively. The normalized objectives are denoted as $\tilde{F}_{cover}$ and $\tilde{F}_{coherence}$. The MOP of GB-SAR deployment is then constructed as follows:

$$\max \ f(x) = \max \left(\tilde{F}_{cover}, \tilde{F}_{coherence} \right). \tag{17}$$

The MOP is decomposed into scalar optimization subproblems ultimately using a linear weighting method to approximate the Pareto front. Each solution obtained from these scalar problems corresponds to an optimal solution. The complete Pareto front of the MOP is constructed by solving all subproblems. The weight coefficients assigned to $\tilde{F}_{\text{cover}}$ and $\tilde{F}_{\text{coherence}}$ are set to be w_1 and w_2, respectively, satisfying $\sum_{i=1}^{2} w_i = 1$. The objective function and associated constraints for the *kth* subproblem are defined as follows:

$$\max f_k \left(x | w_i^k \right) = \max \left(w_1^k \tilde{F}_{\text{cover}} + w_2^k \tilde{F}_{\text{coherence}} \right)$$
$$\text{s.t.} \begin{cases} C_1 : \tilde{F}_{\text{cover}} \geq C_{\text{cover}} \\ C_2 : \tilde{F}_{\text{coherence}} \geq C_{\text{coherence}}. \end{cases}$$

The minimum coherence constraint $C_{\text{coherence}}$ and coverage constraint C_{cover}, respectively, ensure the interferometric measurement accuracy of GB-SAR and the avoidance of excessive monitoring blind spots. Table I lists the key parameters and their values used in the calculation of objective functions.

TABLE I
PARAMETER SETTINGS IN THE CALCULATION OF OBJECTIVE FUNCTIONS

Parameter	Value
$\gamma_{\text{geom}}, \gamma_{\text{DC}}, \gamma_{\text{temp}}, \gamma_{\text{proc}}$	0.99
PN	3.45×10^{-14} W
P_t	10 W
G_t	30 dB
G_r	30 dB
L	1.1
s	1.0 cm
m_v	0.15
$B_\perp$	0.01 m
C_{cover}	50%
$C_{\text{coherence}}$	50%

C. Solving of MOP Based on EQN

This section first models the GB-SAR deployment problem as an MDP and subsequently details the specific architecture of the EQN designed to tackle the high-dimensional challenges.

1) MDP Modeling of the Optimization Process: The sequential decision-making process for deployment is modeled as an MDP tuple $(s, a, s', r_{\text{ext}})$, where s represents the current state, a denotes the action taken by the agent, s' is the next state resulting from executing action a in state s, and r_{ext} is the external reward received after the transition. These tuples are obtained through the interaction between the agent and the environment and stored in the experience replay buffer.

The state of a GB-SAR system containing N devices can be represented as follows:

$$s = \left[(x_1, y_1, z_1, \theta_1), \ldots, (x_N, y_N, z_N, \theta_N) \right]. \tag{18}$$

Each element within this representation denotes the specific state of each device. x_i, y_i, and z_i, respectively, denote the row index, column index, and elevation of the *ith* device within the candidate deployment space. The orientation θ_i is defined as the azimuth angle between the signal transmission direction and true North. Since the state concatenates all device tuples, the same physical deployment could form up to $N!$ different states under different orderings. To remove this redundancy, the tuples are sorted by ascending row index x_i before forming the state, with ties broken by column index y_i. This yields a single ordering for each deployment, so the $N!$ permutations collapse to one state and the Q-network avoids learning redundant mappings.

The agent executes an action based on the current state s to adjust the spatial position and orientation of each device within each step. The action is defined as follows:

$$a = \left[(\Delta x_1, \Delta y_1, \Delta \theta_1), \ldots, (\Delta x_N, \Delta y_N, \Delta \theta_N) \right]. \tag{19}$$

Each element corresponds to the action performed by each device. Δx_i and Δy_i denote the change in position along the row and column directions for the *ith* device, respectively, with $\Delta x_i, \Delta y_i \in \{-\mu, \ldots, 0, \ldots, \mu\}$, where μ is the maximum step size. The orientation change $\Delta \theta_i$ satisfies: $\Delta \theta_i \in \{\alpha \cdot \beta \mid 360 \pmod{\alpha} = 0 \wedge \alpha > 0, \beta \in [0, 360/\alpha - 1] \wedge \beta \in \mathbb{Z}\}$, where α denotes the manually decided azimuthal angular resolution.

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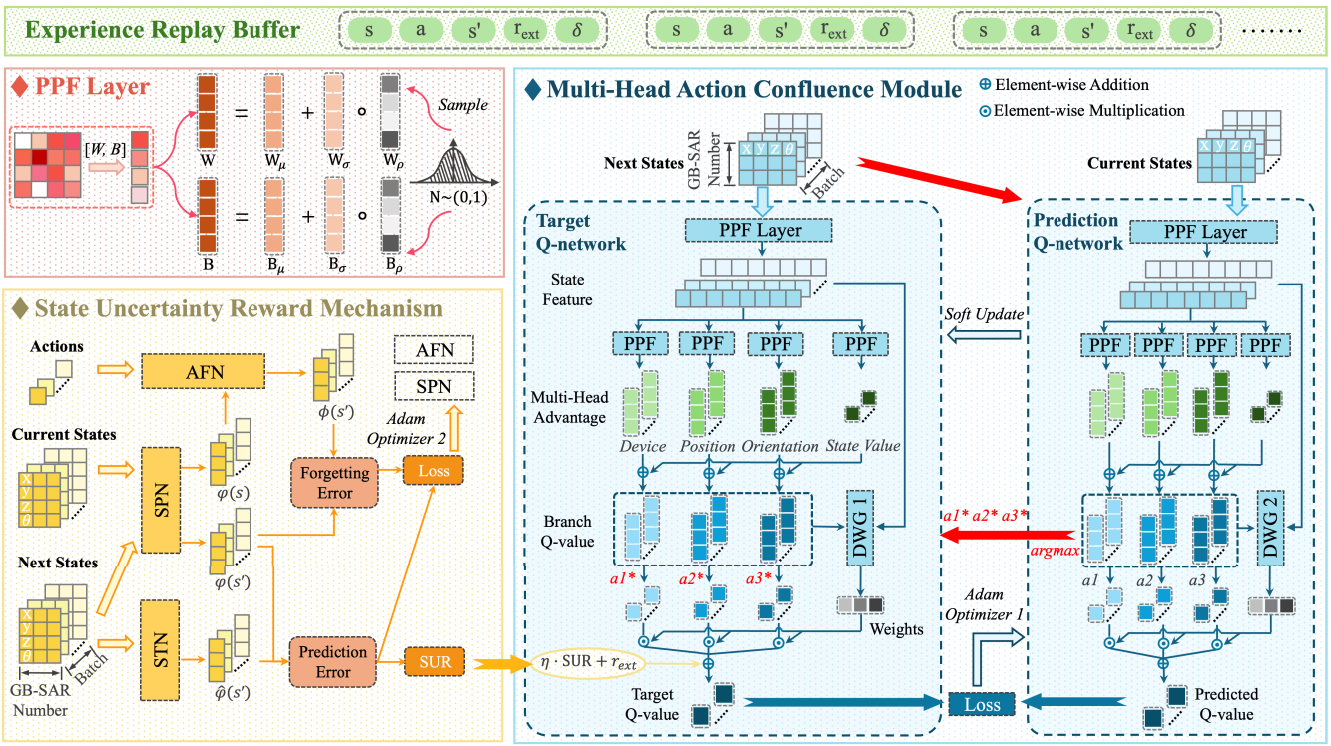


Fig. 3. Overall network architecture of EQN for GB-SAR deployment. It integrates MHAC for dimensionality reduction, PPF for directed exploration, and SUR for intrinsic motivation.

The reward function quantifies the immediate return received by the agent after performing action a in state s during each step. The reward function for the *kth* scalarized subproblem, which also serves as the external reward, is defined as follows:

$$r_{ext} = w_1^k \tilde{F}_{\text{cover}} + w_2^k \tilde{F}_{\text{coherence}} \tag{20}$$

where w_i^k denotes the weight coefficient for the *kth* objective function.

2) Network Architecture: The structure of the EQN is shown in Fig. 3. Experiences acquired during the agent's interaction with the environment are stored in a PER buffer. Batches of experiences are sampled based on their priorities for agent learning. Initially, these sampled experiences $(s, a, s', r_{\text{ext}}, \delta)$ are fed into the MHAC module. It decomposes the calculation of action advantages into multiple parallel branches and produces corresponding subadvantage values instead of directly outputting a single Q value. The DWG then aggregates these subadvantage values, synthesizing an aggregated Q value corresponding to the complete action, which is then used for the loss function calculation. The PPF is embedded within the MHAC module's network. The PPF mechanism generates learnable perturbations at the network level, directionally guiding Q value computation. It guides the agent to perform more directed and broader exploration. The sampled experiences are also fed in parallel into the STN and SPN. These networks evaluate the uncertainty of state prediction and obtain the corresponding SUR for each batch of experiences. The AFN is integrated to mitigate model forgetting concurrently during training. The SUR values are combined with the objective-driven external rewards to jointly compute the final total reward. This total reward subsequently drives the Q-values, steering the agent toward the continuous pursuit of novel states.

3) MHAC Module: The MHAC module addresses the challenges of increasingly action spaces through an action decomposition and aggregation mechanism. High-dimensional composite actions are decomposed into multiple independent subaction heads. This strategy significantly reduces the complexity of the Q value output layer. The DWG learns based on the current state and assigns relative importance weights to each subaction head. This aggregation method assesses the relative importance of different subaction dimensions for the overall decision. It avoids simple averaging or direct concatenation, thereby reducing bias in Q value estimation.

The prediction Q-network and target Q-network share the same architecture and generate predicted Q values and target Q values, respectively, which serve for subsequent loss function calculations. The architecture of the prediction Q-network and target Q-network is shown in Fig. 4. State s is initially processed by a shared neural network backbone that comprises linear layers followed by ReLU activation functions to extract a feature vector f. The extracted features are then fed into two parallel branches: the state value branch and the multihead action advantage branch. The state value branch outputs the state value $V_p(s)$ via linear layers that apply ReLU activation functions. The multihead advantage branch is further split into three independent heads, corresponding to the distinct decision dimensions: device selection, position change, and orientation change. Each head estimates a subadvantage

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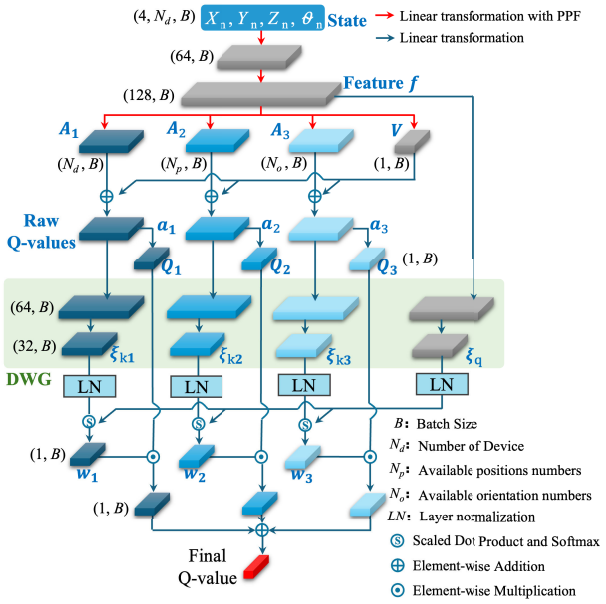


Fig. 4. Architectural Diagram of the prediction Q-network and target Q-network.

$A_{k,p}(s, a_k)(k = 1, 2, 3)$ through ReLU-activated linear layers. a_k represents the specific action taken by the agent in the *kth* subaction dimension and is obtained directly from the experience tuple. The predicted Q value in the *kth* subaction dimension for selecting a_k under s is computed as follows:

$$Q_{k,p}(s, a_k) = V_p(s) + \left[A_{k,p}(s, a_k) - \frac{\sum_{a_k} A_{k,p}(s, a_k)}{N_k}\right] \quad (21)$$

where N_k represents the number of actions in the *kth* subaction dimension.

The weights for each subaction are then computed through the DWG. The feature vector f is projected to a query vector $\xi_{q,p}$. Simultaneously, the raw Q values of each head are projected to key vectors $\xi_{k,p}$. The purpose of these transformations is to map both f and $Q_{k,p}(s)$ into the same-dimensional latent space. To stabilize this computation in early training, each projection is followed by a layer normalization, which rescales $\xi_{q,p}$ and $\xi_{k,p}$ to a common scale. The raw attention scores $S_{k,p}$ for each subaction are obtained via scaled dot product

$$S_{k,p} = \left(\xi_{q,p} \cdot \xi_{k,p}\right) \Big/ \sqrt{D} \quad (22)$$

where D is the dimension of the latent space. The raw dot product may grow large; dividing by $\sqrt{D}$ rescales the scores to a moderate range. These raw attention scores are normalized using a softmax function to produce the final weight vector

$$w_{k,p} = \exp\left(S_{k,p}\right) \Big/ \sum_{k=1}^{3} \exp\left(S_{k,p}\right). \quad (23)$$

The final predicted Q value is obtained by a weighted summation of the Q values for each subaction

$$Q_p(s, a_k) = \sum_{k=1}^{3} w_{k,p} \cdot Q_{k,p}(s, a_k). \quad (24)$$

In early training, the raw Q values are noisy. If left unchecked, an overestimated head could gain an outsized weight and thus amplify its own estimation error, forming a feedback loop. The two mechanisms work together to counter this. The layer normalization bounds the magnitude of each key vector so that an overestimated raw Q value cannot translate its inflated scale directly into a dominant attention score. The $\sqrt{D}$ scaling then keeps the dot-product scores within a moderate range so that the softmax stays responsive and the weights are not driven to collapse onto a single head. In this way, the two mechanisms mitigate the propagation of estimation noise into the weights so that no single head can dominate the aggregated Q value or monopolize the gradient, which helps stabilize early training.

The target Q-network receives s' and the total reward r_{total}. r_{total} is the sum of r_{ext} and the SUR value obtained from the SUR module discussed in Section III-C4. The action used to compute the target Q value is not directly taken from the experience tuple. The agent first feeds s' into the prediction Q-network to generate raw Q values. The actions maximizing their respective raw Q value vectors are defined as $a_k^* = \arg\max_{a_k} Q_{k,p}(s', a_k)$. These specific actions $\{a_k^*\}$ are subsequently used to compute the target Q value. Such a mechanism helps mitigate potential overestimation issues in the target network. The target network parameters are updated using a soft update mechanism during training. Parameters from the prediction network are progressively copied to the target network with a small learning rate τ. This smoothed update approach enhances learning stability. The aggregated Q value from the target network is computed using the same method as the prediction network as follows:

$$Q_t\left(s', a_k^*\right) = \sum_{k=1}^{3} w_k \cdot Q_{k,t}\left(s', a_k^*\right) \quad (25)$$

$$Q_{k,t}\left(s', a_k^*\right) = V_t\left(s'\right) + \left[A_{k,t}\left(s', a_k^*\right) - \frac{\sum_{a_k^*} A_{k,t}\left(s', a_k^*\right)}{N_k}\right]. \quad (26)$$

The final target Q value y is calculated as follows:

$$y = r_{\text{total}} + \gamma_d Q_t\left(s', a_k^*\right) \quad (27)$$

where γ_d is the discount factor. The training objective of EQN is to minimize the temporal difference error (TD error) between the predicted Q value and the target Q value. The loss function can be defined as follows:

$$\mathcal{L}_Q = \mathbb{E}\left\{\left[y - Q_p(s, a)\right]^2\right\}. \quad (28)$$

Furthermore, the priority δ of each experience is defined by the absolute value of the TD error

$$\delta = \left|y - Q_p(s, a)\right|. \quad (29)$$

4) SUR Module: The SUR module consists of three networks: the SPN, STN, and AFN, as shown in Fig. 5. The SPN is a trainable feedforward neural network that receives the current state s and the next state s' and outputs their feature representations $\varphi(s)$ and $\varphi(s')$. The STN shares the same architecture as the SPN, and all its network parameters are entirely frozen after initialization and do not participate

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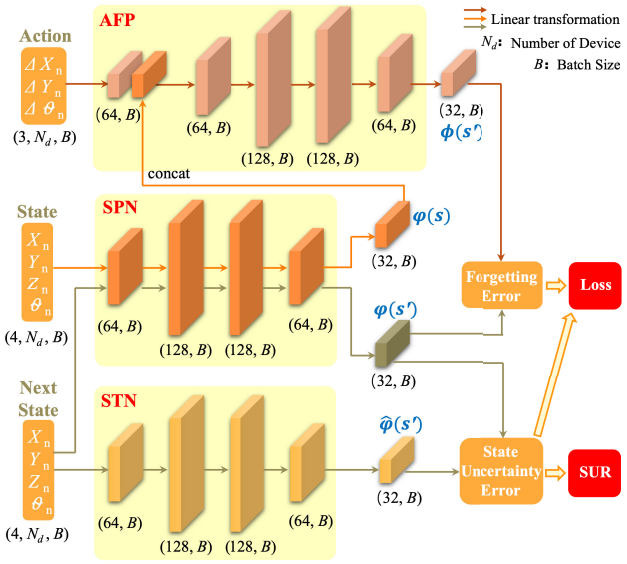


Fig. 5. Diagram of the SUR module.

in any training process. The STN aims to provide a fixed reference feature representation $\hat{\varphi}(s')$ for s'. The AFN receives $\varphi(s)$ and action a and predicts the feature representation $\phi(s')$ of the next state. The raw SUR is denoted as SUR_{init}. State uncertainty refers to the difference between the feature representation $\varphi(s')$ predicted by the SPN and the fixed reference feature $\hat{\varphi}(s')$. A larger uncertainty indicates that the model is unfamiliar with the state. The agent receives a higher intrinsic reward SUR_{init} from this increased discrepancy, encouraging it to prioritize exploration of these unfamiliar states. SUR_{init} is derived from the state uncertainty error e_u, which measures the discrepancy between $\varphi(s')$ and $\hat{\varphi}(s')$. SUR_{init} is defined as follows:

$$SUR_{init} = \mathbb{E}\left(e_u^2\right) \tag{30}$$

$$e_u = \left\|\hat{\varphi}\left(s'\right) - \varphi\left(s'\right)\right\|_2. \tag{31}$$

The Welford algorithm [55] is then employed to estimate the standard deviation σ_r of SUR_{init}, resulting in the normalized SUR, defined as follows:

$$SUR = 1 - \exp\left(-\frac{SUR_{\text{init}}}{\sigma_r}\right). \tag{32}$$

The final total reward, denoted as r_{total}, combines the external reward and the intrinsic reward as follows:

$$r_{\text{total}} = r_{\text{ext}} + \eta \cdot SUR \tag{33}$$

where η is a positive coefficient that balances the contributions of external reward and SUR.

Derived from the uncertainty error e_u, the SUR serves as a meaningful proxy for genuine state novelty. When a state is visited frequently, its related experiences are repeatedly sampled from the replay buffer. The SPN gradually learns its representation, driving $\varphi(s')$ toward the reference $\hat{\varphi}(s')$. The error e_u then converges to low and stable values, decreasing the SUR and diminishing the incentive for the agent to revisit these states. Conversely, for a rarely visited state, the SPN remains insufficiently trained. Its output $\varphi(s')$ acts as an unconstrained extrapolation. In a high-dimensional feature space, it deviates significantly from $\hat{\varphi}(s')$, yielding a large e_u and an elevated SUR that steers the agent toward unfamiliar regions. This mechanism holds even though the STN projects states into an arbitrary feature space, because the STN is a strictly deterministic mapping that consistently generates an identical reference for any given state, providing a fixed anchor for familiarity evaluation.

The SPN and AFN are jointly trained through a common loss function $\mathcal{L}_R$. This loss function minimizes both the state uncertainty error e_u and the forgetting error e_f, defined as follows:

$$\mathcal{L}_R = \mathbb{E}\left(e_u^2\right) + \mathbb{E}\left(e_f^2\right) \tag{34}$$

$$e_f = \left\|\phi\left(s'\right) - \varphi\left(s'\right)\right\|_2. \tag{35}$$

The AFN stabilizes learned representations and mitigates catastrophic forgetting within the SPN. It captures global transition dynamics by predicting the next state feature $\phi(s')$ from the current feature $\varphi(s)$ and the action a. For familiar states, the SPN produces a stable $\varphi(s')$ that the AFN learns to reproduce, so the forgetting error e_f remains small. As training proceeds on other states, the SPN's feature assignments for previously learned states may drift, increasing the forgetting error e_f. Minimizing e_f enforces consistency between the SPN output $\varphi(s')$ and the AFN prediction $\phi(s')$. Since the AFN encodes a global mapping fit over all observed transitions, altering its prediction to match a single drifted state would conflict with the many transitions it has already learned. Its prediction tends to stay near the value learned while the state was well represented, drifting far more slowly than that of the SPN. Consequently, consistency is restored mainly by adjusting the SPN, pulling $\varphi(s')$ back toward the AFN prediction. This mechanism complements e_u. While minimizing e_u drives $\varphi(s')$ toward the fixed reference $\hat{\varphi}(s')$ to generate the novelty signal, minimizing e_f keeps that signal stable against forgetting, preserving the reliability of the SUR.

5) PPF Module: The PPF module injects the factorized learnable perturbation of NoisyNet [21] directly into the Q-network weights and biases, replacing the ϵ-greedy strategy with a directed exploration that steers the agent toward higher reward paths. The PPF module replaces the traditional linear layers with perturbed linear layers in the shared feature extraction layer and in both the state-value and action-advantage branches of the MHAC module. In each perturbed linear layer, the weight W and bias B are jointly defined by a learnable mean (W_μ and B_μ), a learnable standard deviation (W_σ and B_σ), and a perturbation term (W_ρ and B_ρ)

$$W = W_\mu + W_\sigma \odot W_\rho \tag{36}$$

$$B = B_\mu + B_\sigma \odot B_\rho \tag{37}$$

where $\odot$ represents element-wise multiplication, and the means and standard deviations are optimized during training through gradient backpropagation. Following the factorized Gaussian noise of NoisyNet, W_ρ and B_ρ are constructed from two scaled standard-normal vectors through the scaling function $f(x) = \text{sign}(x)\sqrt{|x|}$ and their outer product. The standard deviations are initialized as $\sigma_{init}/(D_{in})^{1/2}$, where σ_{init} is a

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positive constant and D_{in} is the input dimension, to keep the initial exploration moderate. The mean parameters are typically initialized using the Kaiming uniform distribution method [56].

Algorithm 1 EQN Training Algorithm

Input: Env $\mathcal{E}$, number of devices N, objective function weights w_i, max episodes M, max steps T
Output: Optimal parameters for EQN

Initialize parameters $\theta_{\text{pred}}, \theta_{\text{target}}, \theta_{\text{SPN}}, \theta_{\text{STN}}, \theta_{\text{AFN}}$ and replay buffer $\mathcal{D}$
for episode = 1 **to** M **do**
  $s \leftarrow \mathcal{E}$.state_initialization()
  resample noise ξ for PPF in θ_{pred}
  for t = 1 **to** T **do**
    $a \leftarrow \arg\max_a Q_p(s;\xi)$; $(s', r_{\text{ext}}) \leftarrow \mathcal{E}$.execute($a$)
    $SUR_{\text{init}} \leftarrow \mathbb{E}[\|STN(s') - SPN(s')\|^2]$
    $SUR \leftarrow 1 - \exp(-SUR_{\text{init}}/\sigma_r)$
    $r_{\text{total}} \leftarrow r_{\text{ext}} + \eta \cdot SUR$
    $Q_{\text{pred}} \leftarrow \sum_k W_{k,p} \cdot Q_{k,p}(s, a; \xi)$
    $a_k{}^* \leftarrow \arg\max_{a_k} Q_p(s', a_k; \xi)$
    $Q_{\text{target}} \leftarrow r_{\text{total}} + \gamma_d \sum_k W_{k,t} \cdot Q_{k,t}(s', a_k{}^*; \xi)$
    Store $(s, a, r_{\text{ext}}, s', \delta)$ into $\mathcal{D}$ with priority $\delta = |Q_{\text{target}} - Q_{\text{pred}}|$
    if t (mod 5) == 0 **and** $|\mathcal{D}| > 1000$ **then**
      Sample a batch $\{(s^j, a^j, r_{\text{ext}}^j, s'^j, \delta^j)\}_{j=1}^B$ from $\mathcal{D}$ by priority
      Compute $Q_{\text{pred}}^j, Q_{\text{target}}^j$ as above
      $\mathcal{L}_Q = \mathbb{E}[(Q_{\text{target}}^j - Q_{\text{pred}}^j)^2]$
      Update $\theta_{\text{pred}} \leftarrow$ Adam optimizer1($\mathcal{L}_Q$)
      soft update $\theta_{\text{target}} \leftarrow \theta_{\text{pred}}$
      $\varphi(s^j) \leftarrow SPN(s^j)$, $\varphi(s'^j) \leftarrow SPN(s'^j)$
      $\hat{\varphi}(s'^j) \leftarrow STN(s'^j)$, $\phi(s'^j) \leftarrow AFN(\varphi(s^j), a^j)$
      $\mathcal{L}_R = \mathbb{E}\{[\hat{\varphi}(s'^j) - \varphi(s'^j)]^2\}$
      $+\mathbb{E}\{[\phi(s'^j) - \varphi(s'^j)]^2\}$
      update $\theta_{\text{SPN}}, \theta_{\text{AFN}} \leftarrow$ Adam optimizer2($\mathcal{L}_R$)
    end if
    $s \leftarrow s'$
  end for
end for

The PPF module enables the agent to explore more paths in a directional manner. The agent may observe a large TD error when executing an action under a specific policy. This indicates that the agent is exploring along a new path, causing uncertainty in the Q value estimates. Subsequent gradient backpropagation will deliver a strong learning signal to the network parameters associated with that policy, increasing the strength of perturbations within those parameters. The enhanced perturbations will increase the Q value estimates for the corresponding actions in future exploration. The policy tends to select those actions because of their corresponding higher Q values, enabling the agent to achieve biased exploration at the policy level. In contrast, a small TD error indicates that the agent is repeatedly exploring familiar paths and the Q value estimates remain relatively stable. Perturbation strength decreases accordingly. This reduction lowers the Q value estimates of the associated actions and suppresses further exploration of familiar paths. The training procedure of the proposed EQN is detailed in Algorithm 1.

IV. Experiments and Discussion

A. Experimental Environment

The study area covers an open-pit coal mine slope in Ordos, China. To evaluate the long-term robustness of the deployment, the analysis includes data from two periods: June 2024 and June 2025, as shown in Fig. 6. The DSM was obtained through UAV photogrammetry. The morphology of the mine slope had undergone significant changes over time. In Period 1, the northwest and northeast slopes were separated by a middle road, with a slope area of approximately 0.45 km^2. By Period 2, the slopes became contiguous and the mining depth increased as mining operations progressed. The area of the mine slopes expanded to 0.62 km^2. The deployment spaces for the two periods were located on the narrow terrace zone opposite the slope, with areas of 0.37 and 0.34 km^2. Furthermore, the average elevation of the deployment area decreased in Period 2, resulting in altered monitoring LOS conditions.

The agent's movement is discretized into steps of $\{-1, 0, 1\}$, resulting in nine effective movement choices, as illustrated in Fig. 6(c). The orientation adjustment angle is set as 22.5°, and the GB-SAR orientation is restricted to face north to ensure effective observation of the slope. A total of nine discrete orientation angles are included. The state space dimension D_s for N deployed devices is calculated as follows:

$$D_s = \binom{N_g}{N} \times N_o^N = \frac{N_g!}{N!\,(N_g - N)!} \times N_o^N \tag{38}$$

where N_g is the number of grids shown in Fig. 6(d), N_o is the number of possible orientations per device, and N is the number of devices. The state space dimensions reach approximately 2×10^{10} for two devices, 9×10^{13} for three devices, and 4×10^{18} for four devices, using a DSM resolution of 10 m in Period 1. The dimension D_a of the action space is calculated as follows:

$$D_a = N_m \times N_o \times N \tag{39}$$

where N_m is the number of movement step choices. The dimensions of the action space reach 162, 243, and 324 for two, three, and four devices, respectively. The core challenge posed by high-dimensional state-action spaces lies in the exponential growth of the domain dimension for the optimal Q-function that the agent aims to approximate. The dimension of the Q-function's domain is defined as the Cartesian product of D_s and D_a. Specifically, the domain dimension reaches approximately 3.2×10^{12} for two devices, 2.9×10^{15} for three devices, and 1.3×10^{20} for four devices. The agent is required to approximate the optimal Q-function and estimate accurate Q values for an extremely large number of state-action pairs. Such a high-dimensional problem rigorously benchmarks the optimization capabilities of the proposed method.

The GB-SAR configuration and parameter settings used in the simulation are provided in Table II. The experiments were performed on a workstation running Ubuntu, using

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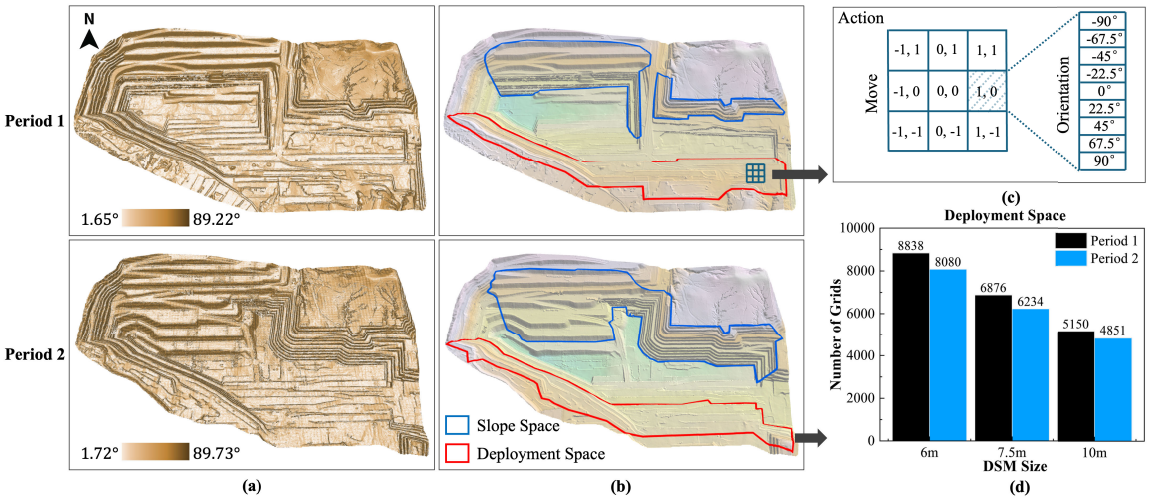


Fig. 6. Study areas in two periods. (a) Slope of the open-pit coal mine. (b) DSM of the open-pit coal mine. (c) Action design of the agent. (d) Number of grids in the deployment space at different DSM resolutions.

TABLE II
PARAMETER SETTINGS IN THE METHOD

Parameter	Value	Description
D_{max}[1]	1500 m	Maximum monitoring range of GB-SAR.
H_{max}[1]	$\pm 120°$	Maximum horizontal scan angle range of GB-SAR.
V_{max}[1]	$\pm 60°$	Maximum vertical scan angle range of GB-SAR.
λ[1]	0.02 m	Radar wavelength of GB-SAR.
τ	0.01	Soft update rate for target network parameters.
γ_d	0.99	Discount factor for target Q-value calculation.
lr_Q	1×10^{-5}	Learning rate for the Q-network optimizer.
lr_{SUR}	1×10^{-5}	Learning rate for SPN and AFN.
$update$	5 steps	Frequency of network parameter updates.
σ_{init}	0.5	Initial standard deviation for perturbation factors.
η	0.525	Scaling coefficient that balances SUR with external reward.
$batch$	64	Batch size sampled from the replay buffer for training.
$buffer$	10^6	Maximum capacity of the prioritized experience replay buffer.
α	0.6	Prioritization exponent for PER.
β	0.4	Initial importance-sampling weight.
$m_{episodes}$	2200	Number of training episodes.
m_{steps}	350	Number of steps per training episode.

[1] These parameters are set with reference to the performance specifications of mainstream commercial FastGBSAR equipment [57], [58].

the PyTorch framework, and equipped with two NVIDIA A100 GPUs.

B. Multiperiod Deployment Experiments in Typical Slope

The environment of the study area is evolving over time, exhibiting different characteristics across periods. Simulations for the two periods validate the robustness of the proposed method under varying temporal and spatial conditions. Experiments were conducted by deploying two, three, and four devices in both periods using a DSM resolution of 7.5 m. To evaluate the tradeoff between monitoring performance and deployment cost across various device numbers in the experiments, F_{cost} is repositioned as a posterior evaluation metric. Once the trained models output the optimal deployment configurations for different N, their overall cost-effectiveness is evaluated by a global indicator, F_{score}, defined as follows:

$$F_{score} = \tilde{F}_{cover} + \tilde{F}_{coherence} - F_{cost} \tag{40}$$

where F_{cost} represents the deployment cost penalty, calculated as $F_{cost} = N \times C$, with N being the number of deployed devices and C the constant unit hardware cost. C is set to 5% to match the typical few-percent monitoring gain from one extra device, thus balancing cost against benefit, and it can be tuned to a mine's budget priorities. Nine weight configurations for objective functions were generated by linearly varying weights from $[0.1, 0.9]$ to $[0.9, 0.1]$ with a step size of 0.1. Each configuration generated an optimal solution through EQN, and these solutions collectively formed the Pareto front. The solution on the Pareto front yielding the minimum Euclidean distance to the ideal point (100, 100) in the objective space is selected as the optimal deployment. It signifies the optimal tradeoff between coverage and coherence. All subsequent experiments were conducted following the aforementioned setup.

Fig. 7(a) shows the Pareto fronts of the optimization results in Period 1. The Pareto front generally shifted toward the upper right (more optimal) region with an increased number of devices, which indicated improvements in both coverage and coherence. The points marked with vertical lines on each frontier in the figure represent the optimal solutions. Coverage of the optimal solution increased from 81.92% to 86.45%, and coherence from 64.87% to 70.71%. The Pareto front in Period 2 also moved toward the upper right, despite changes in slope structure and monitoring conditions, as shown in Fig. 7(b). Coverage of the optimal solution slightly increased from 78.92% to 81.16%, and coherence showed a

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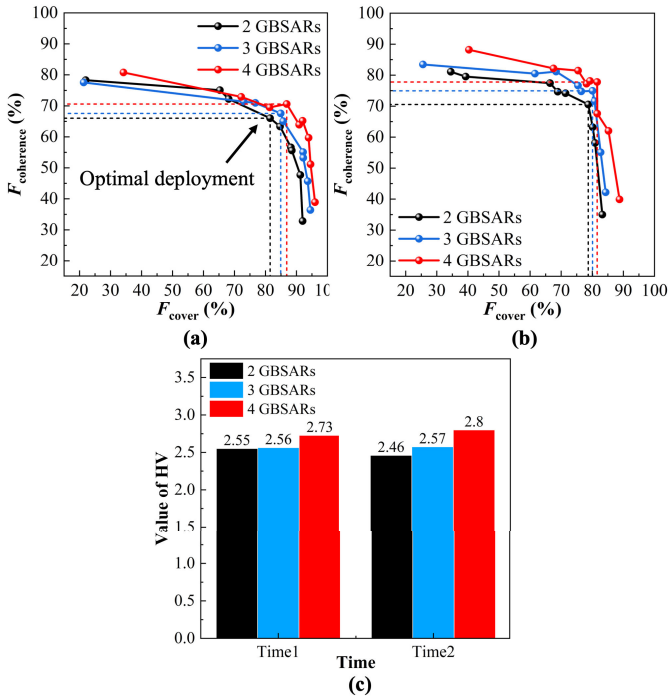


Fig. 7. Pareto fronts and HV values of the optimization results. (a) Period 1. (b) Period 2. (c) HV values.

more substantial improvement, rising from 70.32% to 78.51%. According to (40), the deployment for both periods achieved the highest overall scores with three devices. The scores for the best deployment in Period 1 were 136.79% for two devices, 137.48% for three devices, and 137.16% for four devices. The corresponding scores in Period 2 were 139.24%, 140.04%, and 139.67%. The benefit gained from adding more devices is insufficient to offset the scoring penalty incurred by deployment costs beyond three devices. Furthermore, the study used the hypervolume (HV) index to quantitatively evaluate Pareto front quality, as shown in Fig. 7(c). HV values increased with the number of devices in both periods. These results suggest that EQN effectively enhances both objectives, even in dynamically changing slope environments.

The deployment results for three devices in both periods are shown in Fig. 8. The devices were distributed uniformly in Period 1, which resulted in nearly complete slope coverage, as shown in Fig. 8(a). High coherence was observed across the upper middle and right slope regions. A small, narrow area in the middle slope exhibited relatively lower coherence, which may be attributed to the reduced incidence angle of the GB-SAR observations resulting from the steep local topography. Devices were also uniformly distributed in Period 2, as shown in Fig. 8(b). The deployment achieved high coverage across most of the slope, except for a small flat area on the left side of the slope. This might be caused by the reduced height of the deployment space, which altered the LOS conditions. The coherence significantly improved, with high-coherence regions continuously distributed across the entire slope. It is noted that these deployments achieved desirable monitoring effectiveness in both periods. The exploration count heatmaps for both periods showed that peak regions were distributed widely across the deployment area. This pattern indicates that agents visited numerous candidate locations rather than concentrating on a limited number of spots. The broad exploration allowed the agents to gather diverse environmental information and helped prevent premature convergence to suboptimal solutions. Overall, the deployment derived using EQN demonstrated satisfactory results for the same slope in different periods, confirming the robustness of EQN.

C. Ablation Experiments

1) Ablation Experiments for Different Modules: The study conducted ablation experiments under the condition of deploying three devices on the slope with a 7.5-m DSM resolution in Period 2, to verify the specific contributions of the MHAC module, the SUR module, and the PPF module in enhancing GB-SAR monitoring performance. The D3QN was used as the baseline model. The Pareto fronts of the optimization results are shown in Fig. 9(a). Coverage of the optimal solution for each module combination remained stable and maintained a high level between 78% and 80%. In contrast, coherence exhibited significant differences. The optimal solution achieved the lowest coherence of 61.27% when using MHAC alone, but it increased to 65.57% after incorporating the SUR module. It suggests that the SUR module effectively encouraged the agent to explore more broadly, finding better deployment locations that enhance coherence. The optimal solution's coherence was further improved to 72.71% after introducing the PPF module, which nearly matched the complete EQN's coherence of 74.95%. This improvement highlighted the core role of the PPF module in guiding the agent to discover improved deployments. The optimal solution's coherence decreased to 63.52% when using only the SUR module and the PPF module. The HV value in Fig. 9(b) exhibited a consistent trend. The MHAC module alone achieved an HV value of 2.14, which increased to 2.32 after adding the SUR module, further growing to 2.47 with the PPF module. The HV value dropped to 2.19 when only the SUR and PPF modules were present. This indicates that these two modules could not effectively enhance monitoring performance without the foundational support of the MHAC module.

The deployment results for different module combinations are presented in Fig. 10. The slope areas enclosed by blue boxes in the figures highlight the primary distinctions among these combinations. Fig. 10(a) shows the results when only the MHAC module was utilized. The coherence within the blue-boxed area is lower than in other combinations. The heatmap displays a relatively sparse distribution of peak regions. The introduction of the SUR and PPF modules brought substantial improvements to both monitoring performance and exploration patterns, as depicted in Fig. 10(b) and (c). Coherence within the blue-boxed regions increased significantly, and peak regions of the heatmaps became more broadly distributed. Fig. 10(d) illustrates the deployment when the model lacked the MHAC module. The combination of the SUR and PPF modules showed significantly compromised coherence and exploration capability. In conclusion, the MHAC-only variant

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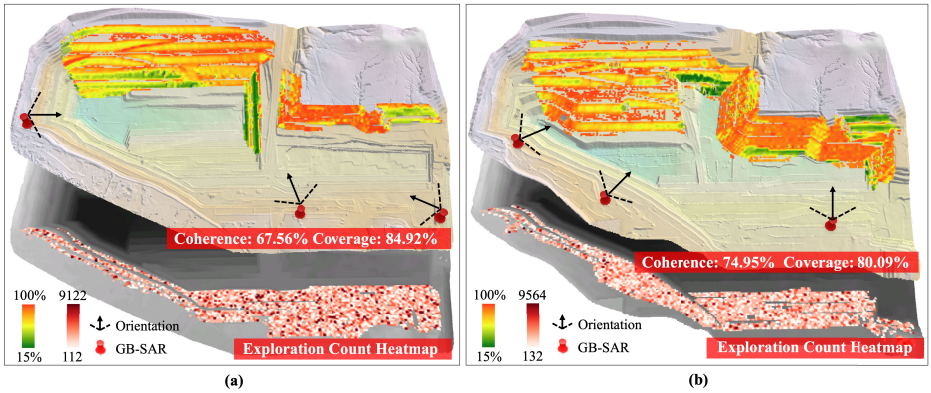


Fig. 8. Deployment results for three devices in both periods. (a) Deployment results for Period 1. (Top) Distribution of coherence, where red regions indicate high-coherence and green regions signify low coherence. (Bottom) Exploration count heatmap, constructed by counting the cumulative visits of the agent to all candidate deployment locations throughout the training phase. It serves to evaluate the spatial exploration breadth of the deployment strategy. Redder areas indicate a higher number of visits. (b) Deployment results for Period 2.

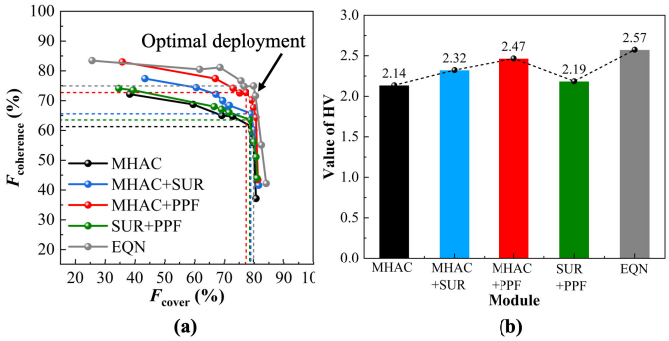


Fig. 9. (a) Pareto fronts and (b) HV values of the optimization results.

TABLE III
EFFECT OF THE MHAC MODULE ON THE COMPLEXITY OF THE ACTION ADVANTAGE BRANCH

Method	Output layer dimension	Action-branch parameters	Convergence episode
With MHAC	21	2,688	1885
Without MHAC	243	31,104	2013

(D3QN + MHAC) attains lower coherence than the complete EQN and exhibits a noticeably narrower exploration pattern. This indicates that the dimensionality reduction of MHAC alone is not sufficient, and without the directed exploration of PPF and the intrinsic motivation of SUR, the agent still settles into local optima. When PPF and SUR are retained, but MHAC is removed, the performance also degrades sharply. Without MHAC's decomposition of the action space, the network converges only to markedly worse solutions in the high-dimensional space, since MHAC markedly reduces the output-layer dimension and parameters of the action advantage branch as shown in Table III. Moreover, adding only SUR or only PPF on top of MHAC still falls short of the complete EQN, confirming that PPF and SUR both contribute rather than either one acting alone. These results demonstrate that the three modules are mutually dependent.

2) Ablation Experiments for Objective Functions: Ablation experiments were conducted to validate the individual contributions of $F_{\text{coherence}}$ and F_{cover} to the optimization performance. All experiments were carried out under the conditions of three devices deployed on the slope with a 7.5-m DSM resolution in Period 2. The agent tended to deploy devices in regions that significantly improved local coherence when optimizing only $F_{\text{coherence}}$. Fig. 11(a) shows that devices clustered noticeably in the left area, with their fields of view concentrated on specific areas of the slope. This deployment achieved the highest coherence of 85.65%, but at the cost of a significant reduction in coverage to 37.63%, leaving large areas of the slope unmonitored. The agent distributed devices uniformly to maximize visibility across the slope when optimizing only F_{cover}, as shown in Fig. 11(b). Coverage reached 81.22%, but coherence sharply decreased to 37.19%, leading to low interferometric measurement quality in most monitored areas. Both heatmaps reveal that exploration breadth was slightly narrower under single-objective optimization than under multiobjective optimization. This limitation occurs because a singular objective restricts the agent's motivation for exploration. The agent only focuses on exploring areas beneficial to a single objective, leading to insufficient exploration of other areas. Overall, these results indicate that both coverage and coherence must be considered simultaneously to ensure the reliability and quality of GB-SAR deployment.

D. Comparative Experiments of Different Deployment Methods

1) Comparison of Deployment Performance: Comparative experiments were conducted among three methods—EQN, D3QN [39], and NSGA-II [29]—under the condition of deploying three devices in Period 2. Experiments were performed at three DSM resolutions: 10, 7.5, and 6 m. The corresponding state space dimensions increased evenly with resolution, enabling testing of each method's adaptability in high-dimensional spaces. The configurations of D3QN and NSGA-II are summarized in Table IV. D3QN is a standard

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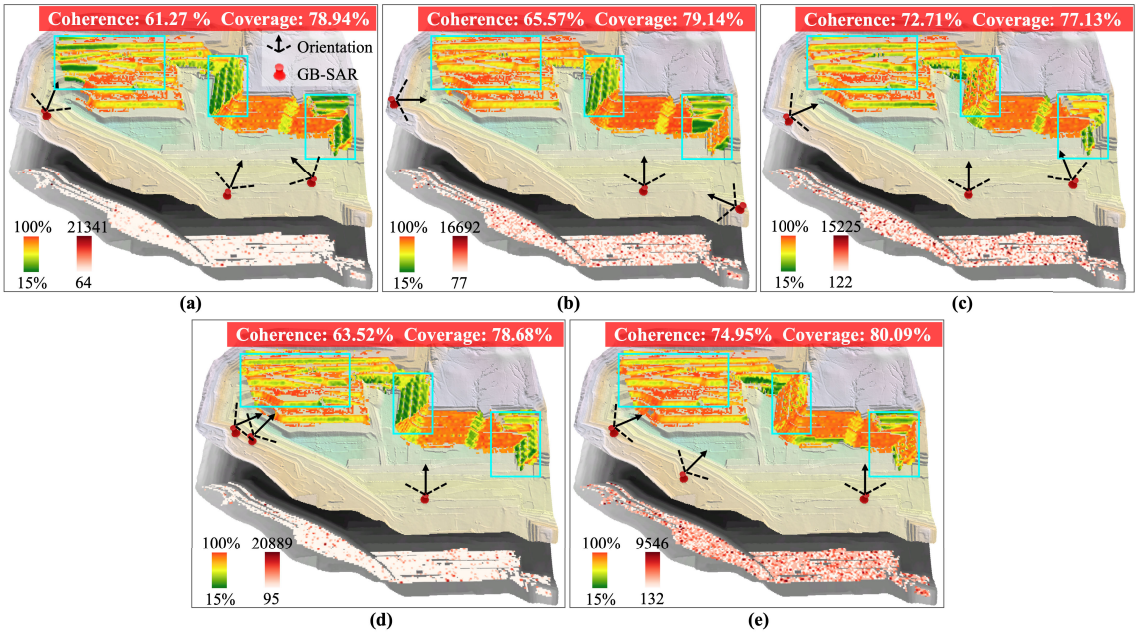


Fig. 10. Deployment results and exploration frequency heatmaps for different module combinations. (a) MHAC only. (b) MHAC with SUR. (c) MHAC with PPF. (d) SUR with PPF. (e) Complete EQN model.

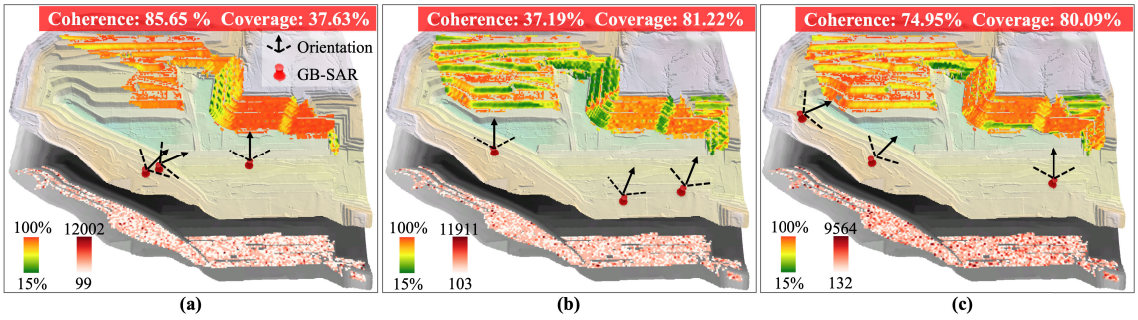


Fig. 11. Deployment results and exploration count heatmaps for different objective functions. (a) Optimizing only $F_{\text{coherence}}$. (b) Optimizing only F_{cover}. (c) Optimizing both $F_{\text{coherence}}$ and F_{cover}.

baseline without the MHAC, PPF, or SUR modules, and both D3QN and NSGA-II adopt their original structures. All methods were tuned to their own optimal settings. The tuning was performed on the 6-m DSM and the optimal settings were reused at the 7.5- and 10-m resolutions, since a configuration that converges stably in the highest dimensional space remains valid for the lower dimensional ones.

The Pareto frontiers at different resolutions are compared in Fig. 12. All methods' frontiers shifted downward and leftward as the resolution increased from 10 to 6 m, reflecting the increased optimization difficulty caused by larger state space dimensions. The results at 10-m resolution show that EQN and D3QN had largely overlapping frontiers and both significantly outperformed NSGA-II. EQN's Pareto frontier began to diverge from D3QN's in coherence when the resolution increased to 7.5 m while maintaining a clear advantage over NSGA-II in both coverage and coherence. The results at 6-m resolution demonstrate an even greater separation of EQN's Pareto frontier from the others in terms of coverage and coherence. Fig. 12(d) presents the HV values for different resolutions. All methods exhibited a decrease in HV with higher resolution; however, EQN's HV decreased at a notably slower rate. EQN consistently achieved higher HV values than the other two methods across all resolutions.

The optimal deployments generated by the three methods at different resolutions are shown in Fig. 13. The results at 10-m resolution indicate that both EQN and D3QN demonstrate excellent performance in coherence and coverage. High-coherence regions are distributed across most areas of the slope, as illustrated in Fig. 13(a) and (b). The deployment using NSGA-II, shown in Fig. 13(c), exhibits notably fewer high-coherence regions alongside numerous monitoring blind spots. EQN's advantages became more apparent when the resolution decreased to 7.5 m. Fig. 13(d) shows that high-coherence regions still covered key slope areas despite a slight overall decline. The D3QN deployment in Fig. 13(e) filled some right-side monitoring blind spots but introduced new low-coherence areas, while NSGA-II's deployment in

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TABLE IV
CONFIGURATIONS OF D3QN AND NSGA-II IN THE COMPARATIVE EXPERIMENTS

Method	Param.	Value	Description
D3QN	lr_Q	5×10^{-5}	Learning rate of the Q-network optimizer.
	$batch$	64	Batch size sampled from the buffer.
	τ	0.01	Soft update rate for the target network.
	γ_d	0.99	Discount factor for the target Q-value.
	α	0.6	Prioritization exponent for PER.
	β_0	0.4	Initial importance-sampling weight.
	$buffer$	10^6	Maximum capacity of the replay buffer.
	ϵ	$1.0 \rightarrow 0.05$	ϵ-greedy rate, linearly annealed.
	Ep.×steps	2200×350	Total training length.
NSGA-II	N_p	350	Number of individuals per generation.
	G	2200	Number of evolutionary generations.
	p_c	0.9	Simulated binary crossover (SBX) probability.
	η_c	20	SBX distribution index.
	p_m	$1/L$	Polynomial mutation probability (L: variables).
	η_m	20	Polynomial mutation distribution index.
	k	2	Binary tournament selection size.

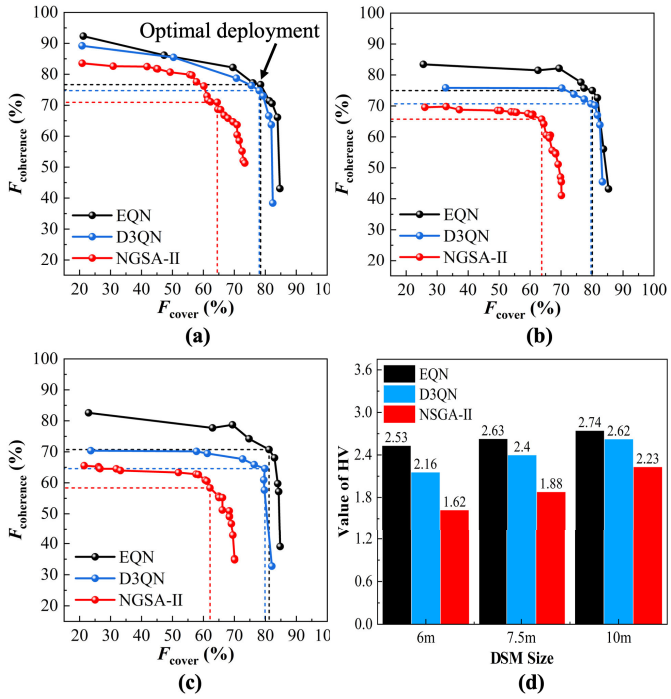


Fig. 12. Pareto fronts and HV values of the optimization results. (a) 10-m resolution results for EQN, D3QN, and NSGA-II. (b) 7.5-m resolution results. (c) 6-m resolution results. (d) HV values at different resolutions.

Fig. 13(f) suffered from an increased number of low-coherence regions and blind spots. The results at 6-m resolution in Fig. 13(g) and (h) show that both EQN and D3QN deployments exhibited low-coherence regions concentrated in the slope's central part, but EQN attained better coherence on the left and far right parts of the slope. NSGA-II's deployment exhibited further expanded blind spots and a significant decline in coherence. While the deployment results above illustrate the spatial distribution of coherence qualitatively, the cumulative distribution function (cdf) of the coherence values is further compared in Fig. 14 to quantify the overall improvement, where a curve closer to the lower right corner reflects a larger proportion of high-coherence pixels. At every resolution, EQN's cdf lies consistently to the lower right of those of D3QN and NSGA-II, and the separation widens as the resolution increases, which statistically confirms that EQN expands the proportion of high-coherence pixels across all test cases. In summary, the relatively higher coherence and coverage achieved by EQN on the slope indicate that the agent can make more precise action selections. This precision benefits from the MHAC module, which reduces gradient interference between different subactions, enabling more stable and accurate parameter updates and thus more accurate Q value estimations of actions. This is attributed to the significantly reduced output parameters of each subaction head, enabling them to receive more focused learning signals.

TABLE V
COMPARISON OF COMPUTATIONAL EFFICIENCY OF DIFFERENT METHODS

Methods	Training Time	Inference Time
EQN	1 h 35 min 54 s	1 min 14 s
D3QN	50 min 34 s	39 s
NSGA-II	1 h 7 min 15 s	—

Furthermore, EQN's exploration heatmaps across all resolutions reveal widespread and dense peaks, which indicates broad and diverse exploration. D3QN's heatmaps display relatively sparser peaks with a considerably larger disparity in overall visit counts; the disparity reaches approximately 30 000 visits at 6-m resolution, roughly twice that of EQN. NSGA-II's exploration heatmaps appear fragmented, with many deployment areas left unexplored. The broader and denser exploration pattern of EQN's heatmaps strongly suggests the effective role of the SUR and PPF modules. They enable the agent to avoid premature convergence and thoroughly explore the deployment space.

Overall, these results demonstrate that as the dimensions of the deployment space grow, EQN consistently exhibits superior adaptability in coherence, coverage, and exploration breadth. This advantage largely stems from the synergistic effects of the MHAC, SUR, and PPF modules, which are absent in the compared methods.

2) Comparison of Computational Efficiency: Table V compares the computational efficiency under Period 2 with a 7.5-m DSM resolution and three devices. EQN and D3QN are trained for 2200 episodes of 350 steps, while NSGA-II evolves a population of 350 over 2200 generations. For EQN and D3QN, the training phase concurrently optimizes nine models that correspond to the different weight configurations. The reported time reflects the total wall-clock duration required for all nine parallel models to reach convergence. As a heuristic method, NSGA-II must restart the evolutionary optimization for every new task. EQN and D3QN leverage one-time offline training for minute-level online inference. Although EQN

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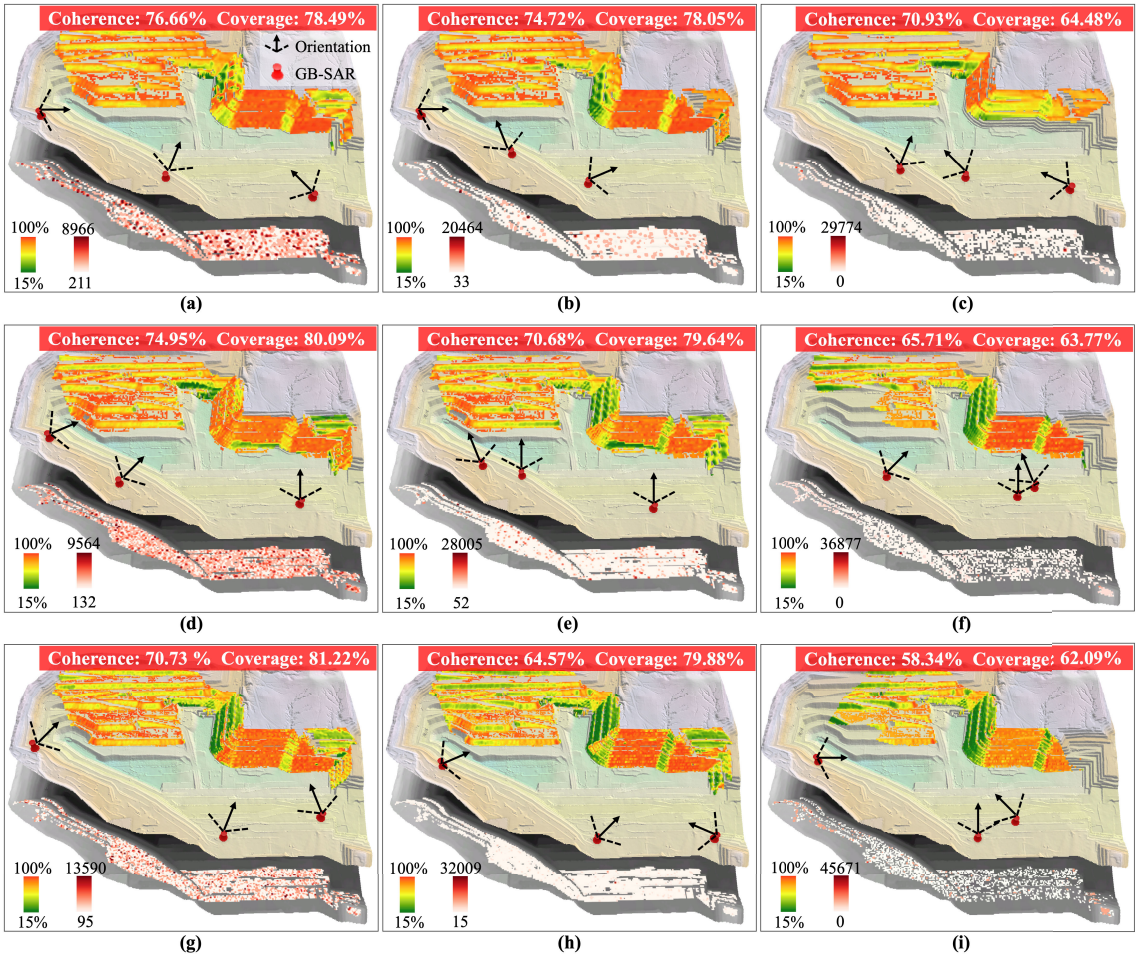


Fig. 13. Deployment results and exploration count heatmaps of EQN, D3QN, and NSGA-II at different DSM resolutions. (a)–(c) 10-m resolution results for EQN, D3QN, and NSGA-II, respectively. (d)–(f) 7.5-m resolution results. (g)–(i) 6-m resolution results.

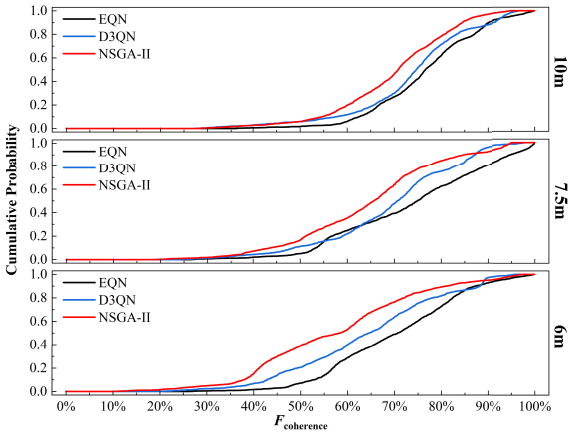


Fig. 14. Coherence cdf curves of EQN, D3QN, and NSGA-II at the 10-, 7.5-, and 6-m DSM resolutions.

trains longer than D3QN due to the additional SUR, PPF, and DWG computations, its inference speed remains comparable while delivering higher coherence and coverage. This computational overhead is an acceptable tradeoff for the performance achieved by EQN.

3) Interferometric and Deformation Analysis of Different Deployments: To evaluate the final monitoring products, the interferograms and deformation maps are simulated and compared in Fig. 15. The interferometric phase is simulated by wrapping the sum of the deformation phase ϕ_{def} and noise phase ϕ_{noise} into $(-\pi, \pi]$ [59]. ϕ_{def} is derived from the true deformation d_{true} as $\phi_{\text{def}} = 4\pi/\lambda_{\text{true}}$ [60]. ϕ_{noise} is sampled from a zero-mean Gaussian distribution whose standard deviation is governed by the per-grid coherence [61]. To retrieve the final deformation maps, the wrapped interferograms are first denoised by a complex-domain Gaussian filter [62] and then unwrapped via a minimum-cost-flow algorithm [63]. The unwrapped phase ϕ_{unw} is then converted to LOS displacement via $d_{\text{LOS}} = \lambda/4\pi\phi_{\text{unw}}$ and finally projected along the slope-aspect direction.

The experiments are conducted under the same setting of Period 2 with three devices at a resolution of 7.5 m. The interferogram obtained by EQN contains the lowest level of decorrelation noise. In contrast, the interferograms produced by the D3QN and NSGA-II models become progressively noisier, with the result of NSGA-II being the most degraded. Although all three methods correctly retrieve the prominent red deformation in the southern part, the discrepancies among

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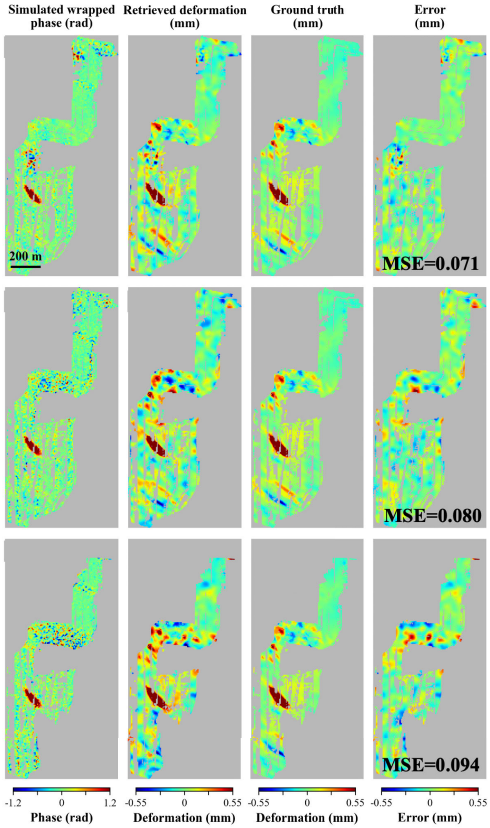


Fig. 15. Simulated wrapped interferograms, retrieved deformation, ground truth, and error maps of EQN, D3QN, and NSGA-II at the typical 7.5-m DSM resolution.

them arise primarily in the surrounding low-coherence zones, where decorrelation noise is easily misidentified as physical deformation. The deformation map retrieved by the EQN maintains the highest consistency with the ground truth, attaining the lowest mean squared error (mse) of 0.071, compared to 0.080 for the D3QN model and 0.094 for NSGA-II. Ultimately, these results confirm that the deployment strategy learned by the EQN method effectively mitigates noise within the interferograms and minimizes deformation retrieval errors, explicitly demonstrating the robustness of this approach in solving high-dimensional deployment problems.

E. Experiments Under Different Action and Exploration Parameters

In each of the following experiments, one parameter is varied, while the others are kept at their default values, so that the individual effect of each parameter on the performance of EQN can be isolated.

1) Experiments With Different Movement Step Sizes and Orientation Adjustment Angles: The effect of the movement step size and the orientation adjustment angle on the performance of EQN is investigated under Period 2 with a 7.5-m DSM resolution and three devices, and the results are listed in Table VI.

A small movement step delivers a better deployment without incurring extra training cost. As the step increases from 1 to 3, the performance rises slightly and then decreases steadily as it further grows to 5 and 7, since a larger step makes the agent skip candidate positions and renders the deployment coarser. The training time stays essentially constant across all step sizes. The exploration heatmap variance decreases as the step grows, so a larger step spreads the agent's visits more widely across the deployment space, yet this broader exploration does not offset the coarser deployment it produces. Although the step of 3 is marginally ahead, this margin is negligible and is specific to the present setting. The step is set to 1 for robustness, as it keeps every candidate position reachable at the finest spatial resolution and thereby avoids missing the potential optimum that a larger step may skip. Conversely, a finer orientation angle trades a higher training cost for a better deployment. Reducing the orientation angle from 30° to 22.5° significantly improves deployment quality by enabling more precise observation directions. However, further reductions yield diminishing returns. Unlike the movement step, a finer

TABLE VI
PERFORMANCE OF EQN UNDER DIFFERENT ACTION AND EXPLORATION PARAMETERS

Parameter	Value	$\tilde{F}_{coherence}$	$\tilde{F}_{cover}$	F_{score}	HV	Heatmap Variance[1]	Training Time
Movement Step Sizes	1	74.95%	80.09%	140.04%	2.63	640.4	1h 35m 54s
	3	75.41%	80.11%	140.52%	2.65	625.2	1h 36m 0s
	5	73.82%	79.20%	138.02%	2.58	595.6	1h 35m 48s
	7	73.14%	78.15%	136.29%	2.54	552.1	1h 36m 6s
Orientation Adjustment Angles	11.25°	75.08%	80.89%	140.97%	2.67	642.0	1h 49m 30s
	15°	75.16%	80.22%	140.38%	2.66	639.0	1h 44m 36s
	22.5°	74.95%	80.09%	140.04%	2.63	640.4	1h 35m 54s
	30°	73.21%	78.52%	136.73%	2.50	638.5	1h 26m 36s
SUR Scaling Coefficient η	0.025	72.16%	78.27%	135.43%	2.45	684.0	1h 36m 18s
	0.275	74.51%	80.12%	139.63%	2.57	652.0	1h 35m 42s
	0.525	74.95%	80.09%	140.04%	2.63	640.4	1h 35m 54s
	0.775	73.69%	79.94%	138.63%	2.55	631.0	1h 36m 6s
PPF Initial Standard Deviation σ_{init}	0.1	69.50%	75.80%	130.30%	2.30	715.0	1h 36m 12s
	0.3	72.30%	77.80%	135.10%	2.46	670.0	1h 35m 36s
	0.5	74.95%	80.09%	140.04%	2.63	640.4	1h 35m 54s
	0.7	71.85%	78.10%	134.95%	2.45	618.0	1h 35m 30s

[1] Heatmap Variance is computed from the agent's visit counts over all candidate deployment locations, a lower value indicates broader exploration.

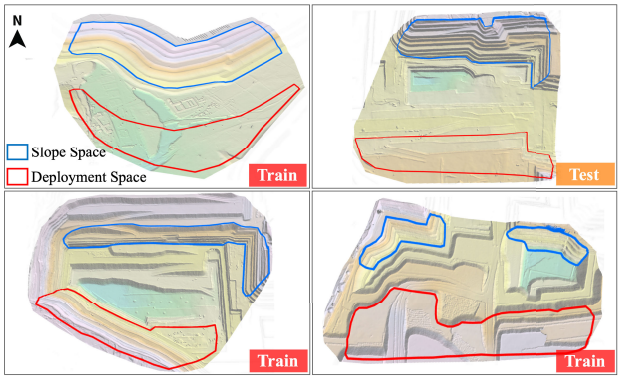


Fig. 16. Four open-pit slope sites used in the generalization experiment. (Top left, bottom left, and bottom right) Sites 1–3 are used for training, and (top right) Site 4 is the test slope.

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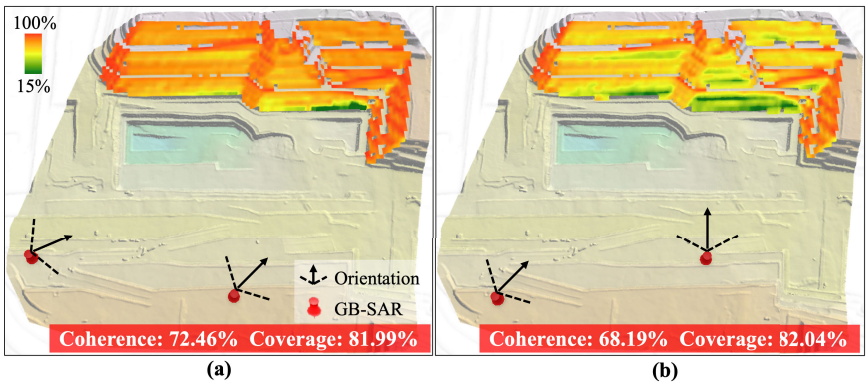


Fig. 17. Deployment results on the test slope. (a) EQN trained from scratch. (b) EQN transferred from Sites 1 to 3 and applied to the test slope by inference only.

angle enlarges the action space and model parameters, which steadily increases the required training time. The exploration heatmap variance stays nearly constant across all angles, so the angle has little effect on the breadth of exploration. The angle is therefore set to 22.5°, which keeps the performance high while avoiding the extra training cost of finer angles.

2) Experiments With Different Exploration Coefficients η and σ_{Init}: The exploration coefficients η and σ_{init} are further analyzed under the Period 2 setting with a 7.5-m DSM resolution and three devices. The results are detailed in Table VI.

The performance exhibits an inverted-U relationship with η, with $\eta = 0.525$ lying in its upper region. As η increases, the performance of EQN first improves and then declines once η becomes too large. A small η provides relatively insufficient intrinsic reward and leads to underexploration, whereas an excessively large η prioritizes novelty at the expense of the external objective, so the broader exploration is no longer converted into better deployments. Similarly, σ_{init} produces a sharper inverted-U relationship that peaks at $\sigma_{init} = 0.5$, with performance degrading at both smaller and larger values. The drop of F_{score} at the small σ_{init} is markedly larger than the largest drop observed when η is varied, indicating EQN's higher sensitivity to σ_{init}. At a large σ_{init}, the strong perturbation broadens exploration further but disturbs the policy stability rather than improving performance, as reflected by the consistent decline across all performance metrics.

These findings highlight the necessary balance between exploration and exploitation. Extreme parameter values typically disrupt this balance, causing either premature convergence or degraded primary task performance. Adopting the defaults of $\eta = 0.525$ and $\sigma_{init} = 0.5$ generally promotes broad exploration while maintaining stable policy convergence.

F. Generalization Experiment

To verify the generalization capability of EQN, an experiment is conducted on four morphologically distinct open-pit slope sites, as shown in Fig. 16. Each training site is additionally rotated by 180° for augmentation, yielding six training scenes for sequential training. The training algorithm selects the optimal model on the Pareto front for each scene, defined as the solution closest to the ideal point, and applies its converged parameters to initialize the subsequent scene. The resulting model proceeds directly to the test slope by inference only. A control EQN model establishes a baseline by training from scratch directly on the same test slope following the identical pipeline. The relatively small footprint of these sites dictates that all six training scenes and the test scene consistently deploy two GB-SAR devices at a 7.5-m DSM resolution. All remaining parameters strictly adhere to the settings presented in Table II.

The optimal deployments on the test slope are compared in Fig. 17. The from-scratch model attains 72.46% coherence and 81.99% coverage, while the transferred model yields 68.19% and 82.04%, respectively. Both models achieve essentially identical coverage, but the transferred policy exhibits a 4.27% drop in coherence, manifesting visually as more low-coherence patches in the central and right sections of the slope. The limited training data primarily cause this performance gap. As a practical constraint, only three base sites are available, lacking the representational capacity to fully cover the diversity of slope morphologies, deployment geometries, and LOS conditions characterizing open-pit mines. Nevertheless, by inference alone, the transferred model still reaches a coverage on par with the from-scratch baseline and a coherence reasonably close to it. It should be noted that when high monitoring accuracy is required, the converged parameters are better employed as a warm start for a brief fine-tuning on the new terrain rather than for direct inference, which still remains far cheaper than training from scratch. These findings represent evidence derived from a single held-out site and exactly three base terrains. The current results therefore serve as an encouraging indication of transferability. Future work will require an assessment across a larger and more diverse set of test slopes. The steady accumulation of additional training sites is expected to systematically narrow this transfer gap.

V. Conclusion

The EQN method proposed in this study mitigates the issue of existing multiobjective optimization methods getting stuck in local optima in high-dimensional spaces. The MHAC module decomposes high-dimensional actions into subaction heads

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and dynamically aggregates their advantage values, ensuring concentrated learning signals and accurate Q value estimation. The PPF module injects NoisyNet-style learnable perturbations into the Q-network, enabling the agent to directionally explore more unfamiliar paths. The SUR module generates intrinsic rewards by quantifying the uncertainty of state predictions. The agent is encouraged to actively explore unfamiliar states, and the catastrophic forgetting problem is mitigated through an AFN. Multiperiod experiments conducted in open-pit coal mines demonstrated the robustness of EQN. Module ablation experiments demonstrated the foundational role of the MHAC module in EQN and the synergistic effect of integrating the SUR and PPF modules on model performance. Objective function ablation experiments emphasized the necessity of simultaneously optimizing coverage and coherence. Comparative experiments with NSGA-II and D3QN proved EQN's stronger adaptability in high-dimensional environments. The sensitivity analysis of the action and exploration parameters clarified the applicability boundaries of the method. Furthermore, the generalization experiment across morphologically distinct open-pit slopes verified the transferability of the learned policy.

A primary limitation lies in the scattering model adopted in the coherence objective, which is calibrated only for incidence angles between 10° and 70°. On sloped terrain, angles exceeding this range cause uncertain extrapolation. Nevertheless, since the coherence objective serves as a relative ranking criterion rather than an absolute measurement, such deviations affect the candidate deployments in a broadly similar manner, so their influence on the relative ranking of the optimized configuration is expected to be limited. Adopting physically based models for wider angular ranges requires additional field calibration and computation, which is reserved for future work. Another direction is to improve the state representation method. While our current lightweight encoding of device positions and orientations accelerates network convergence, there is still potential to better capture the overall features of the deployment space. Future approaches could leverage graph neural networks, treating devices as nodes and topological relationships as edges to learn richer spatial features.

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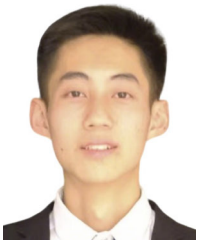
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